Binary Suffix Learning in Non-Binary Bases — Full Overnight Experiment

Task: Given n encoded as base-B digits, predict $n \bmod 2^p$.

This notebook runs the full (B, p) experimental grid with:

- Architecture matching Charton & Narayanan (2025): 4-layer encoder, 1-layer decoder, dim 512, 8 heads
- On-the-fly data generation (no stored datasets)
- Checkpoint saving after every experiment for Colab disconnect resilience
- Per-class accuracy breakdown to detect step-like learning patterns
- Linear probing of hidden representations

Estimated runtime: 6–12 hours on Colab Pro (T4/V100/A100).

0. Mount Google Drive for Persistent Checkpoints

In [3]:
```
import os

# Mount Google Drive so checkpoints survive disconnects
try:
    from google.colab import drive
    drive.mount('/content/drive')
    SAVE_DIR = '/content/drive/MyDrive/binary_suffix_experiment'
    IN_COLAB = True
except ImportError:
    SAVE_DIR = './binary_suffix_experiment'
    IN_COLAB = False

os.makedirs(SAVE_DIR, exist_ok=True)
print(f"Checkpoint directory: {SAVE_DIR}")
print(f"Running in Colab: {IN_COLAB}")
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.
mount("/content/drive", force_remount=True).
Checkpoint directory: /content/drive/MyDrive/binary_suffix_experiment
Running in Colab: True
```

1. Imports & Device Setup

In [4]:
```
import math
import time
import json
import pickle
from collections import defaultdict
from pathlib import Path
```

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader, IterableDataset

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {device}')
if device.type == 'cuda':
    print(f'GPU: {torch.cuda.get_device_name(0)}')
    print(f'Memory: {torch.cuda.get_device_properties(0).total_memory / 1e9:.1f} GB
print(f'PyTorch: {torch.__version__}')
```

```
Device: cuda
GPU: NVIDIA L4
Memory: 23.7 GB
PyTorch: 2.10.0+cu128
```

2. Configuration

All experiment parameters in one place. Edit `BASES` and `P_VALUES` to adjust the grid.

In [5]:
```
# ── Reproducibility ──────────────────────────────────────────────────────
GLOBAL_SEED = 1337
np.random.seed(GLOBAL_SEED)
torch.manual_seed(GLOBAL_SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(GLOBAL_SEED)

# ── Run controls ─────────────────────────────────────────────────────────
RUN_SWEEP = True
RUN_ANALYSIS = True
RUN_PROBES = True

# On reruns, skip experiments whose saved accuracy is already good enough.
SKIP_SUCCESSFUL_ON_SWEEP = True

# Mark a run as "successful / learned" once it reaches this accuracy.
SUCCESS_ACC_THRESHOLD = 0.90

# Only allow patience-based early stopping AFTER the run has already crossed
# this threshold. This prevents long flat plateaus from getting cut off.
EARLY_STOP_ACC_THRESHOLD = 0.98

# Optional: force specific experiments to rerun even if they already succeeded.
# Example: FORCE_RERUN = {(3, 4), (5, 6)}
FORCE_RERUN = set()

# If False, unsuccessful runs resume from their saved checkpoint/history.
# If True, unsuccessful runs start over from scratch.
RETRAIN_FROM_SCRATCH = False
```

```
# ── Experimental grid ──────────────────────────────────────────────────
# Full grid from project outline:
# BASES = [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 16, 24, 27, 32]
# P_VALUES = [1, 2, 3, 4, 5, 6, 7, 8]

# First pass reduced grid for quick iteration:
BASES = [2, 3, 5, 8, 10, 16, 27]
P_VALUES = [1, 2, 4, 6, 8]

# ── Architecture (matching Charton & Narayanan) ────────────────────────
D_MODEL = 512
N_HEADS = 8
ENC_LAYERS = 4
#DEC_LAYERS = 1
DIM_FF = 4 * D_MODEL  # 2048
DROPOUT = 0.1

# ── Training ───────────────────────────────────────────────────────────
N_MAX = 10**8            # sample n from [1, N_MAX]
EPOCH_SIZE = 300_000     # examples per epoch (matches Charton's convention)
N_EPOCHS = 40            # 40 epochs = 12M examples per experiment
BATCH_SIZE = 1024
LR = 3e-4
WARMUP_STEPS = 500
EVAL_SIZE = 50_000       # test set size per evaluation

# ── Checkpointing / early stopping ─────────────────────────────────────
SAVE_EVERY = 1
MIN_EPOCHS = 5
EARLY_STOPPING_PATIENCE = 4
MIN_DELTA = 1e-3

# ── Logging ────────────────────────────────────────────────────────────
LOG_INTERVAL = 200       # print loss every N batches
EVAL_EVERY = 1           # evaluate every N epochs

print(f'Grid: {len(BASES)} bases × {len(P_VALUES)} p-values = {len(BASES)*len(P_VAL
print(f'Architecture: enc={ENC_LAYERS}L, d={D_MODEL}, h={N_HEADS}')
print(f'Training: up to {N_EPOCHS} epochs × {EPOCH_SIZE:,} examples = {N_EPOCHS*EPO
print(f'Success threshold: {SUCCESS_ACC_THRESHOLD:.2f}')
print(f'Early-stop threshold: {EARLY_STOP_ACC_THRESHOLD:.2f}')
```

```
Grid: 7 bases × 5 p-values = 35 experiments
Architecture: enc=4L, d=512, h=8
Training: up to 40 epochs × 300,000 examples = 12.0M examples per experiment
Success threshold: 0.90
Early-stop threshold: 0.98
```

3. Data Generation (On-the-Fly)

We use an `IterableDataset` that generates data on the fly, so we never store datasets. This matches the Charton approach of training on freshly generated random examples.

In [6]:
```
def int_to_base_digits(n: int, base: int) -> list:
    """Convert n to base-B digits, MSD first."""
    if n == 0:
        return [0]
    digits = []
    while n > 0:
        digits.append(n % base)
        n //= base
    return digits[::-1]


def max_seq_len(n_max: int, base: int) -> int:
    """Maximum number of digits for any n <= n_max in given base."""
    return math.ceil(math.log(n_max + 1) / math.log(base)) + 1

class BinarySuffixDataset(IterableDataset):
    """
    Infinite streaming dataset for the binary suffix task.

    Token convention (matching Charton's Int2Int):
      0 = EOS
      1 = BOS  (we prepend this to encoder input)
      2..B+1 = digits 0..B-1
      B+2 = PAD

    Encoder input:  [BOS, d_L+2, d_{L-1}+2, ..., d_0+2, EOS, PAD, ...]
    Decoder target:  class label in {0, ..., 2^p - 1}

    We use classification (encoder-only effective) rather than seq2seq for
    the target, but keep the encoder input format compatible with Int2Int.
    """
    def __init__(self, base, p, n_max, max_len, seed=0):
        self.base = base
        self.p = p
        self.mod = 2 ** p
        self.n_max = n_max
        self.max_len = max_len  # includes BOS + EOS
        self.pad_id = base + 2
        self.bos_id = 1
        self.eos_id = 0
        self.seed = seed

    def __iter__(self):
        worker_info = torch.utils.data.get_worker_info()
        seed = self.seed
        if worker_info is not None:
            seed += worker_info.id
        rng = np.random.default_rng(seed)

        while True:
            n = int(rng.integers(1, self.n_max + 1))
            digits = int_to_base_digits(n, self.base)
            tokens = [self.bos_id] + [d + 2 for d in digits] + [self.eos_id]
            tokens = tokens[:self.max_len]
            tokens = tokens + [self.pad_id] * (self.max_len - len(tokens))
```

```
            label = n % self.mod

            yield (
                torch.tensor(tokens, dtype=torch.long),
                torch.tensor(label, dtype=torch.long),
                torch.tensor(n, dtype=torch.long),
            )

def make_eval_data(base, p, n_max, size, max_len, seed=99):
    """Generate a fixed evaluation set (returned as tensors)."""
    rng = np.random.default_rng(seed)
    ns = rng.integers(1, n_max + 1, size=size)
    mod = 2 ** p
    labels = ns % mod
    pad_id = base + 2

    all_tokens = []
    for n in ns:
        digits = int_to_base_digits(int(n), base)
        tokens = [1] + [d + 2 for d in digits] + [0]  # BOS ... EOS
        tokens = tokens[:max_len]
        tokens = tokens + [pad_id] * (max_len - len(tokens))
        all_tokens.append(tokens)

    X = torch.tensor(all_tokens, dtype=torch.long)
    y = torch.tensor(labels.astype(np.int64), dtype=torch.long)
    n_tensor = torch.tensor(ns.astype(np.int64), dtype=torch.long)
    return X, y, n_tensor

# Quick test
msl = max_seq_len(N_MAX, 3)
print(f'Max seq len for base 3, N_MAX={N_MAX:.0e}: {msl} (+ BOS/EOS = {msl+2})')
```

```
Max seq len for base 3, N_MAX=1e+08: 18 (+ BOS/EOS = 20)
```

4. Model Architecture

Encoder-decoder transformer matching the Charton & Narayanan setup:

- Encoder: 4 layers, dim 512, 8 heads, bidirectional
- Decoder: 1 layer, dim 512, 8 heads, with cross-attention

However, since our output is a *classification* (one of 2^p residue classes), we use the encoder to produce a representation and add a classification head. This is architecturally equivalent to an encoder-only model but keeps the encoder structure identical to Charton's for comparability. We also provide an optional seq2seq mode.

For the classification head, we mean-pool over non-pad encoder outputs.

In [7]:
```
class BinarySuffixTransformer(nn.Module):
    """
    Encoder-only classifier matching Charton's encoder architecture.
```

```
    Architecture:
      - Learned token + positional embeddings
      - N-layer TransformerEncoder (pre-norm)
      - Mean-pool over non-pad positions
      - Linear classification head -> 2^p classes
    """
    def __init__(self, vocab_size, num_classes, d_model=512, nhead=8,
                 num_layers=4, dim_feedforward=2048, max_len=64,
                 dropout=0.1, pad_id=None):
        super().__init__()
        self.d_model = d_model
        self.pad_id = pad_id

        self.token_embed = nn.Embedding(vocab_size, d_model, padding_idx=pad_id)
        self.pos_embed = nn.Embedding(max_len, d_model)
        self.embed_dropout = nn.Dropout(dropout)
        self.embed_scale = math.sqrt(d_model)

        encoder_layer = nn.TransformerEncoderLayer(
            d_model=d_model,
            nhead=nhead,
            dim_feedforward=dim_feedforward,
            dropout=dropout,
            batch_first=True,
            norm_first=True,
            activation='gelu',
        )
        self.encoder = nn.TransformerEncoder(
            encoder_layer, num_layers=num_layers,
            enable_nested_tensor=False,
        )
        self.layer_norm = nn.LayerNorm(d_model)
        self.classifier = nn.Linear(d_model, num_classes)

        self._init_weights()

    def _init_weights(self):
        for p in self.parameters():
            if p.dim() > 1:
                nn.init.xavier_uniform_(p)

    def forward(self, x):
        B, L = x.shape
        positions = torch.arange(L, device=x.device).unsqueeze(0)

        h = self.token_embed(x) * self.embed_scale + self.pos_embed(positions)
        h = self.embed_dropout(h)

        # Padding mask
        pad_mask = (x == self.pad_id) if self.pad_id is not None else None

        h = self.encoder(h, src_key_padding_mask=pad_mask)

        # Mean-pool over non-pad positions
        if pad_mask is not None:
            mask = (~pad_mask).unsqueeze(-1).float()  # (B, L, 1)
```

```
            h = (h * mask).sum(dim=1) / mask.sum(dim=1).clamp(min=1)
        else:
            h = h.mean(dim=1)

        h = self.layer_norm(h)
        return self.classifier(h)

    def get_layer_representations(self, x):
        """Return hidden states after each encoder layer (for probing)."""
        B, L = x.shape
        positions = torch.arange(L, device=x.device).unsqueeze(0)
        h = self.token_embed(x) * self.embed_scale + self.pos_embed(positions)

        pad_mask = (x == self.pad_id) if self.pad_id is not None else None

        if pad_mask is not None:
            mask = (~pad_mask).unsqueeze(-1).float()
        else:
            mask = torch.ones(B, L, 1, device=x.device)

        representations = []
        # Collect after each layer
        for layer in self.encoder.layers:
            h = layer(h, src_key_padding_mask=pad_mask)
            pooled = (h * mask).sum(dim=1) / mask.sum(dim=1).clamp(min=1)
            representations.append(pooled.detach())

        return representations

def count_parameters(model):
    return sum(p.numel() for p in model.parameters() if p.requires_grad)

# Quick param count check
_test_model = BinarySuffixTransformer(
    vocab_size=35, num_classes=256, d_model=D_MODEL, nhead=N_HEADS,
    num_layers=ENC_LAYERS, dim_feedforward=DIM_FF, max_len=30, pad_id=34
)
print(f'Test model params: {count_parameters(_test_model):,}')
del _test_model
```

```
Test model params: 12,775,168
```

5. Training & Evaluation Infrastructure

Includes:

- Learning rate warmup (linear) then constant
- Per-class accuracy breakdown
- Step-like learning curve detection
- Checkpointing to Google Drive

In [8]:
```
def get_lr(step, warmup_steps, base_lr):
    """Linear warmup then constant LR."""
    if step < warmup_steps:
```

```
        return base_lr * (step + 1) / warmup_steps
    return base_lr

@torch.no_grad()
def evaluate_detailed(model, X_eval, y_eval, batch_size=2048):
    """Evaluate with per-class accuracy breakdown."""
    model.eval()
    num_classes = int(y_eval.max().item()) + 1
    correct_per_class = torch.zeros(num_classes, dtype=torch.long)
    total_per_class = torch.zeros(num_classes, dtype=torch.long)
    all_correct = 0
    all_total = 0

    for i in range(0, len(X_eval), batch_size):
        x = X_eval[i:i+batch_size].to(device)
        y = y_eval[i:i+batch_size].to(device)
        preds = model(x).argmax(dim=-1)
        correct = (preds == y)

        all_correct += correct.sum().item()
        all_total += y.size(0)

        for c in range(num_classes):
            mask_c = (y == c)
            correct_per_class[c] += (correct & mask_c).sum().item()
            total_per_class[c] += mask_c.sum().item()

    overall_acc = all_correct / all_total
    class_acc = {}
    for c in range(num_classes):
        if total_per_class[c] > 0:
            class_acc[c] = correct_per_class[c].item() / total_per_class[c].item()
        else:
            class_acc[c] = float('nan')

    return overall_acc, class_acc

@torch.no_grad()
def evaluate_by_residue_mod_2p(model, X_eval, y_eval, ns_eval, p_check, batch_size=
    """
    Evaluate accuracy grouped by n mod 2^p_check.
    This tests whether the model has learned to classify by binary suffixes,
    mirroring Section 4.1 of Charton & Narayanan.
    """
    model.eval()
    mod_check = 2 ** p_check
    correct_by_residue = defaultdict(int)
    total_by_residue = defaultdict(int)

    for i in range(0, len(X_eval), batch_size):
        x = X_eval[i:i+batch_size].to(device)
        y = y_eval[i:i+batch_size].to(device)
        ns = ns_eval[i:i+batch_size]
        preds = model(x).argmax(dim=-1)
        correct = (preds == y).cpu()
        residues = (ns % mod_check).numpy()
```

```
        for j in range(len(ns)):
            r = int(residues[j])
            correct_by_residue[r] += int(correct[j].item())
            total_by_residue[r] += 1

    acc_by_residue = {}
    for r in sorted(total_by_residue.keys()):
        if total_by_residue[r] > 0:
            acc_by_residue[r] = correct_by_residue[r] / total_by_residue[r]

    return acc_by_residue

def save_checkpoint(state, filepath):
    """Save checkpoint to disk."""
    torch.save(state, filepath)

def load_checkpoint(filepath):
    """Load checkpoint from disk."""
    if os.path.exists(filepath):
        return torch.load(filepath, map_location=device, weights_only=False)
    return None
```

6. Main Training Function

In [9]:

```
def get_result_score(result):
    """Best available scalar score for deciding success / skipping."""
    if result is None:
        return float('-inf')
    candidates = []
    for key in ('best_acc', 'final_acc'):
        val = result.get(key, None)
        if isinstance(val, (int, float)):
            candidates.append(float(val))
    return max(candidates) if candidates else float('-inf')

def experiment_succeeded(result, threshold=SUCCESS_ACC_THRESHOLD):
    return result.get('completed', False) and get_result_score(result) >= threshold

def clear_experiment_artifacts(exp_name, save_dir):
    """Delete saved state so an experiment can restart from scratch."""
    for suffix in ['_checkpoint.pt', '_results.json', '_best_model.pt', '_model.pt'
        path = os.path.join(save_dir, f'{exp_name}{suffix}')
        if os.path.exists(path):
            os.remove(path)

def train_experiment(base, p, save_dir=SAVE_DIR, resume=True):
    """
    Train a single (base, p) experiment with full logging and checkpointing.

    Returns a results dict with training history, per-class accuracy, etc.
    Saves checkpoints after each epoch to `save_dir`.

    Rerun policy:
```

```
    - Skip completed runs whose saved score already exceeds SUCCESS_ACC_THRESHOLD
    - Resume incomplete / unsuccessful runs from checkpoint by default.
    - If RETRAIN_FROM_SCRATCH=True, unsuccessful runs are restarted from scratch.
    """

    exp_name = f'B{base}_p{p}'
    ckpt_path = os.path.join(save_dir, f'{exp_name}_checkpoint.pt')
    result_path = os.path.join(save_dir, f'{exp_name}_results.json')
    best_model_path = os.path.join(save_dir, f'{exp_name}_best_model.pt')
    model_path = os.path.join(save_dir, f'{exp_name}_model.pt')

    existing = None
    if os.path.exists(result_path):
        with open(result_path, 'r') as f:
            existing = json.load(f)

    if (base, p) in FORCE_RERUN:
        print(f'  [{exp_name}] FORCE_RERUN set; rerunning regardless of saved resul
        if RETRAIN_FROM_SCRATCH:
            clear_experiment_artifacts(exp_name, save_dir)
            existing = None

    elif existing is not None and SKIP_SUCCESSFUL_ON_SWEEP and experiment_succeeded
        print(f'  [{exp_name}] Already successful (score={get_result_score(existing
        return existing

    elif existing is not None:
        old_score = get_result_score(existing)
        if RETRAIN_FROM_SCRATCH:
            print(f'  [{exp_name}] Existing run below threshold (score={old_score:.
            clear_experiment_artifacts(exp_name, save_dir)
            existing = None
        else:
            print(f'  [{exp_name}] Existing run below threshold (score={old_score:.

    num_classes = 2 ** p
    msl = max_seq_len(N_MAX, base) + 2  # +2 for BOS and EOS
    vocab_size = base + 3  # digits 0..B-1 -> tokens 2..B+1, plus BOS(1), EOS(0), P
    pad_id = base + 2

    # Build model
    model = BinarySuffixTransformer(
        vocab_size=vocab_size,
        num_classes=num_classes,
        d_model=D_MODEL,
        nhead=N_HEADS,
        num_layers=ENC_LAYERS,
        dim_feedforward=DIM_FF,
        max_len=msl,
        dropout=DROPOUT,
        pad_id=pad_id,
    ).to(device)

    optimizer = torch.optim.Adam(model.parameters(), lr=LR, betas=(0.9, 0.98), eps=
    criterion = nn.CrossEntropyLoss()
```

```
    # Generate fixed eval set
    X_eval, y_eval, ns_eval = make_eval_data(base, p, N_MAX, EVAL_SIZE, msl, seed=1

    # History
    history = {
        'epoch': [], 'train_loss': [], 'test_acc': [],
        'class_acc': [], 'examples_seen': [],
    }
    start_epoch = 0
    global_step = 0

    # Resume from checkpoint if available
    if resume:
        ckpt = load_checkpoint(ckpt_path)
        if ckpt is not None:
            model.load_state_dict(ckpt['model_state'])
            optimizer.load_state_dict(ckpt['optimizer_state'])
            history = ckpt['history']
            start_epoch = ckpt['epoch'] + 1
            global_step = ckpt['global_step']
            print(f'  [{exp_name}] Resumed from epoch {start_epoch}')

    nparams = count_parameters(model)
    chance = 1.0 / num_classes
    print(f'  [{exp_name}] classes={num_classes} | max_len={msl} | params={nparams:

    # Training data loader (streaming)
    exp_seed = GLOBAL_SEED + 1000 * base + p
    train_ds = BinarySuffixDataset(base, p, N_MAX, msl, seed=exp_seed)

    num_workers = 2 if device.type == 'cuda' else 0
    train_loader = DataLoader(
        train_ds,
        batch_size=BATCH_SIZE,
        num_workers=num_workers,
        pin_memory=(device.type == 'cuda'),
        persistent_workers=(num_workers > 0),
    )

    # Use AMP for speed on GPU
    use_amp = (device.type == 'cuda')
    scaler = torch.amp.GradScaler('cuda', enabled=use_amp)

    best_acc = max(history['test_acc']) if history['test_acc'] else -1.0
    epochs_without_improve = 0
    best_model_path = os.path.join(save_dir, f'{exp_name}_best_model.pt')
    completed_epochs = start_epoch

    for epoch in range(start_epoch, N_EPOCHS):
        model.train()
        epoch_loss = 0.0
        epoch_correct = 0
        epoch_total = 0
        t0 = time.time()

        for x, y, n_val in train_loader:
```

```
            x = x.to(device, non_blocking=True)
            y = y.to(device, non_blocking=True)

            lr = get_lr(global_step, WARMUP_STEPS, LR)
            for pg in optimizer.param_groups:
                pg['lr'] = lr

            optimizer.zero_grad(set_to_none=True)
            with torch.amp.autocast('cuda', enabled=use_amp):
                logits = model(x)
                loss = criterion(logits, y)

            scaler.scale(loss).backward()
            scaler.unscale_(optimizer)
            torch.nn.utils.clip_grad_norm_(model.parameters(), 5.0)
            scaler.step(optimizer)
            scaler.update()

            epoch_loss += loss.item() * y.size(0)
            epoch_correct += (logits.argmax(-1) == y).sum().item()
            epoch_total += y.size(0)
            global_step += 1

            if epoch_total >= EPOCH_SIZE:
                break

        avg_loss = epoch_loss / max(epoch_total, 1)
        train_acc = epoch_correct / max(epoch_total, 1)

        test_acc, class_acc = evaluate_detailed(model, X_eval, y_eval)

        elapsed = time.time() - t0
        completed_epochs = epoch + 1

        history['epoch'].append(epoch)
        history['train_loss'].append(avg_loss)
        history['test_acc'].append(test_acc)
        history['class_acc'].append({str(k): v for k, v in class_acc.items()})
        history['examples_seen'].append(
            (history['examples_seen'][-1] if history['examples_seen'] else 0) + epo
        )

        print(
            f'    epoch {epoch+1:>3}/{N_EPOCHS}  '
            f'loss={avg_loss:.4f}  train={train_acc:.4f}  test={test_acc:.4f}  '
            f'best={best_acc:.4f}  '
            f'({elapsed:.0f}s, {epoch_total/max(elapsed,1e-6):.0f} ex/s)'
        )

        # Save checkpoint every epoch
        if (epoch + 1) % SAVE_EVERY == 0:
            save_checkpoint({
                'epoch': epoch,
                'global_step': global_step,
                'model_state': model.state_dict(),
                'optimizer_state': optimizer.state_dict(),
```

```
                'history': history,
                'base': base,
                'p': p,
            }, ckpt_path)

        # Track best model
        if test_acc > best_acc + MIN_DELTA:
            best_acc = test_acc
            epochs_without_improve = 0
            torch.save(model.state_dict(), best_model_path)
        else:
            # Only start counting patience once the model is already good enough.
            if best_acc >= EARLY_STOP_ACC_THRESHOLD:
                epochs_without_improve += 1
            else:
                epochs_without_improve = 0

        if (
            completed_epochs >= MIN_EPOCHS
            and best_acc >= EARLY_STOP_ACC_THRESHOLD
            and epochs_without_improve >= EARLY_STOPPING_PATIENCE
        ):
            print(
                f'    early stopping triggered at epoch {completed_epochs} '
                f'(best_acc={best_acc:.4f} >= {EARLY_STOP_ACC_THRESHOLD:.2f})'
            )
            break

    # Ensure best model is loaded for final evaluation
    if os.path.exists(best_model_path):
        model.load_state_dict(torch.load(best_model_path, map_location=device))

    # Final evaluation with residue breakdown
    final_acc, final_class_acc = evaluate_detailed(model, X_eval, y_eval)

    # Residue analysis for interesting cases
    residue_analysis = {}
    if p >= 2:
        for p_check in range(1, min(p + 2, 9)):
            acc_by_res = evaluate_by_residue_mod_2p(model, X_eval, y_eval, ns_eval,
            residue_analysis[f'mod_2^{p_check}'] = acc_by_res

    # Save final model weights separately
    # model_path = os.path.join(save_dir, f'{exp_name}_model.pt')
    torch.save(model.state_dict(), model_path)

    # Compile results
    success = best_acc >= SUCCESS_ACC_THRESHOLD
    results = {
        'base': base, 'p': p, 'num_classes': num_classes,
        'final_acc': final_acc,
        'final_class_acc': {str(k): v for k, v in final_class_acc.items()},
        'best_acc': best_acc,
        'success': success,
        'success_threshold': SUCCESS_ACC_THRESHOLD,
        'early_stop_threshold': EARLY_STOP_ACC_THRESHOLD,
```

```
        'epochs_completed': completed_epochs,
        'history': history,
        'residue_analysis': {k: {str(kk): vv for kk, vv in v.items()}
                             for k, v in residue_analysis.items()},
        'params': nparams,
        'completed': True,
        'nu_2_B': nu_2(base),
    }

    with open(result_path, 'w') as f:
        json.dump(results, f, indent=2)

    status = 'SUCCESS' if success else 'NEEDS RETRY'
    print(f'  [{exp_name}] Done. Final acc = {final_acc:.4f}, best acc = {best_acc:
    return results

def nu_2(n):
    """2-adic valuation of n."""
    if n == 0:
        return float('inf')
    v = 0
    while n % 2 == 0:
        v += 1
        n //= 2
    return v
```

7. Run the Full Experimental Grid

Experiments are run in order of expected difficulty (power-of-2 bases first, then even, then odd). Each experiment saves results independently, so if Colab disconnects, re-running this cell will skip completed experiments automatically.

In [10]:
```
# Sort bases: powers of 2 first, then even, then odd
def base_sort_key(b):
    v = nu_2(b)
    if b & (b - 1) == 0:  # power of 2
        return (0, -v, b)
    elif b % 2 == 0:
        return (1, -v, b)
    else:
        return (2, 0, b)

def load_saved_results(bases, p_values, save_dir=SAVE_DIR):
    saved = {}
    for base in bases:
        for p in p_values:
            result_path = os.path.join(save_dir, f'B{base}_p{p}_results.json')
            if os.path.exists(result_path):
                with open(result_path, 'r') as f:
                    saved[(base, p)] = json.load(f)
    return saved

def summarize_pending_experiments(bases, p_values, save_dir=SAVE_DIR):
```

```
    saved = load_saved_results(bases, p_values, save_dir)
    pending, successful = [], []
    for base in bases:
        for p in p_values:
            key = (base, p)
            if key in FORCE_RERUN:
                pending.append(key)
            elif key in saved and experiment_succeeded(saved[key], SUCCESS_ACC_THRE
                successful.append(key)
            else:
                pending.append(key)
    return saved, successful, pending

sorted_bases = sorted(BASES, key=base_sort_key)
print(f'Base order: {sorted_bases}')
print(f'P values: {P_VALUES}')
print(f'Total experiments in grid: {len(sorted_bases) * len(P_VALUES)}')

saved_results, successful_experiments, pending_experiments = summarize_pending_expe
print(f'Already successful (skip): {len(successful_experiments)}')
print(f'Pending / retrain: {len(pending_experiments)}')
if pending_experiments:
    print('Pending list:', pending_experiments)

all_results = dict(saved_results)
total_start = time.time()

for base, p in pending_experiments:
    exp_key = (base, p)
    print(f'\n{"="*70}')
    print(f'Experiment: base={base}, p={p} (n mod {2**p}), v₂(B)={nu_2(base)}')
    print(f'{"="*70}')

    try:
        result = train_experiment(base, p)
        all_results[exp_key] = result
    except Exception as e:
        print(f'  ERROR: {e}')
        import traceback; traceback.print_exc()
        all_results[exp_key] = {'error': str(e), 'base': base, 'p': p}

    # Save master results after each experiment
    master_path = os.path.join(SAVE_DIR, 'all_results_summary.json')
    summary = {}
    for (b, pp), r in all_results.items():
        summary[f'B{b}_p{pp}'] = {
            'base': b, 'p': pp,
            'final_acc': r.get('final_acc', None),
            'best_acc': r.get('best_acc', None),
            'score': get_result_score(r),
            'success': experiment_succeeded(r, SUCCESS_ACC_THRESHOLD),
            'nu_2': nu_2(b),
            'completed': r.get('completed', False),
        }
    with open(master_path, 'w') as f:
        json.dump(summary, f, indent=2)
```

```
total_elapsed = time.time() - total_start
print(f'\nSweep finished in {total_elapsed/3600:.2f} hours')
print(f'Successful runs now: {sum(experiment_succeeded(r, SUCCESS_ACC_THRESHOLD) fo
```

```
Base order: [16, 8, 2, 10, 3, 5, 27]
P values: [1, 2, 4, 6, 8]
Total experiments in grid: 35
Already successful (skip): 21
Pending / retrain: 14
Pending list: [(10, 6), (10, 8), (3, 2), (3, 4), (3, 6), (3, 8), (5, 2), (5, 4), (5,
6), (5, 8), (27, 2), (27, 4), (27, 6), (27, 8)]

======================================================================
Experiment: base=10, p=6 (n mod 64), v₂(B)=1
======================================================================
  [B10_p6] Existing run below threshold (score=0.2401); resuming / extending trainin
g.
  [B10_p6] Resumed from epoch 40
  [B10_p6] classes=64 | max_len=12 | params=12,656,192 | chance=0.0156
  [B10_p6] Done. Final acc = 0.2401, best acc = 0.2404  [NEEDS RETRY]

======================================================================
Experiment: base=10, p=8 (n mod 256), v₂(B)=1
======================================================================
  [B10_p8] Existing run below threshold (score=0.0615); resuming / extending trainin
g.
  [B10_p8] Resumed from epoch 40
  [B10_p8] classes=256 | max_len=12 | params=12,754,688 | chance=0.0039
  [B10_p8] Done. Final acc = 0.0607, best acc = 0.0615  [NEEDS RETRY]

======================================================================
Experiment: base=3, p=2 (n mod 4), v₂(B)=0
======================================================================
  [B3_p2] Existing run below threshold (score=0.2548); resuming / extending trainin
g.
  [B3_p2] Resumed from epoch 40
  [B3_p2] classes=4 | max_len=20 | params=12,625,924 | chance=0.2500
  [B3_p2] Done. Final acc = 0.2544, best acc = 0.2548  [NEEDS RETRY]

======================================================================
Experiment: base=3, p=4 (n mod 16), v₂(B)=0
======================================================================
  [B3_p4] Existing run below threshold (score=0.0686); resuming / extending trainin
g.
  [B3_p4] Resumed from epoch 40
  [B3_p4] classes=16 | max_len=20 | params=12,632,080 | chance=0.0625
  [B3_p4] Done. Final acc = 0.0686, best acc = 0.0686  [NEEDS RETRY]

======================================================================
Experiment: base=3, p=6 (n mod 64), v₂(B)=0
======================================================================
  [B3_p6] Existing run below threshold (score=0.0170); resuming / extending trainin
g.
  [B3_p6] Resumed from epoch 40
  [B3_p6] classes=64 | max_len=20 | params=12,656,704 | chance=0.0156
  [B3_p6] Done. Final acc = 0.0160, best acc = 0.0170  [NEEDS RETRY]

======================================================================
Experiment: base=3, p=8 (n mod 256), v₂(B)=0
======================================================================
```

```
  [B3_p8] Existing run below threshold (score=0.0043); resuming / extending trainin
g.
  [B3_p8] Resumed from epoch 40
  [B3_p8] classes=256 | max_len=20 | params=12,755,200 | chance=0.0039
  [B3_p8] Done. Final acc = 0.0034, best acc = 0.0043  [NEEDS RETRY]

======================================================================
Experiment: base=5, p=2 (n mod 4), v₂(B)=0
======================================================================
  [B5_p2] Existing run below threshold (score=0.5060); resuming / extending trainin
g.
  [B5_p2] Resumed from epoch 40
  [B5_p2] classes=4 | max_len=15 | params=12,624,388 | chance=0.2500
  [B5_p2] Done. Final acc = 0.5060, best acc = 0.5060  [NEEDS RETRY]

======================================================================
Experiment: base=5, p=4 (n mod 16), v₂(B)=0
======================================================================
  [B5_p4] Existing run below threshold (score=0.1270); resuming / extending trainin
g.
  [B5_p4] Resumed from epoch 40
  [B5_p4] classes=16 | max_len=15 | params=12,630,544 | chance=0.0625
  [B5_p4] Done. Final acc = 0.1270, best acc = 0.1277  [NEEDS RETRY]

======================================================================
Experiment: base=5, p=6 (n mod 64), v₂(B)=0
======================================================================
  [B5_p6] Existing run below threshold (score=0.0323); resuming / extending trainin
g.
  [B5_p6] Resumed from epoch 40
  [B5_p6] classes=64 | max_len=15 | params=12,655,168 | chance=0.0156
  [B5_p6] Done. Final acc = 0.0323, best acc = 0.0323  [NEEDS RETRY]

======================================================================
Experiment: base=5, p=8 (n mod 256), v₂(B)=0
======================================================================
  [B5_p8] Existing run below threshold (score=0.0041); resuming / extending trainin
g.
  [B5_p8] Resumed from epoch 40
  [B5_p8] classes=256 | max_len=15 | params=12,753,664 | chance=0.0039
  [B5_p8] Done. Final acc = 0.0041, best acc = 0.0041  [NEEDS RETRY]

======================================================================
Experiment: base=27, p=2 (n mod 4), v₂(B)=0
======================================================================
  [B27_p2] Existing run below threshold (score=0.5046); resuming / extending trainin
g.
  [B27_p2] Resumed from epoch 29
  [B27_p2] classes=4 | max_len=9 | params=12,632,580 | chance=0.2500
    epoch  30/40  loss=0.6939  train=0.5079  test=0.5018  best=0.5046  (18s, 16924 e
x/s)
    epoch  31/40  loss=0.6941  train=0.5096  test=0.4997  best=0.5046  (17s, 17528 e
x/s)
    epoch  32/40  loss=0.6937  train=0.5098  test=0.5018  best=0.5046  (17s, 17506 e
x/s)
    epoch  33/40  loss=0.6937  train=0.5103  test=0.5009  best=0.5046  (17s, 17603 e
```

```
x/s)
    epoch  34/40  loss=0.6933  train=0.5103  test=0.4992  best=0.5046  (17s, 17624 e
x/s)
    epoch  35/40  loss=0.6935  train=0.5102  test=0.5028  best=0.5046  (17s, 17600 e
x/s)
    epoch  36/40  loss=0.6936  train=0.5120  test=0.5025  best=0.5046  (17s, 17646 e
x/s)
    epoch  37/40  loss=0.6932  train=0.5118  test=0.5000  best=0.5046  (17s, 17590 e
x/s)
    epoch  38/40  loss=0.6936  train=0.5118  test=0.5008  best=0.5046  (17s, 17608 e
x/s)
    epoch  39/40  loss=0.6933  train=0.5119  test=0.5003  best=0.5046  (17s, 17633 e
x/s)
    epoch  40/40  loss=0.6931  train=0.5122  test=0.4994  best=0.5046  (17s, 17621 e
x/s)
  [B27_p2] Done. Final acc = 0.5046, best acc = 0.5046  [NEEDS RETRY]

======================================================================
Experiment: base=27, p=4 (n mod 16), v₂(B)=0
======================================================================
  [B27_p4] Existing run below threshold (score=0.1258); resuming / extending trainin
g.
  [B27_p4] Resumed from epoch 5
  [B27_p4] classes=16 | max_len=9 | params=12,638,736 | chance=0.0625
    epoch   6/40  loss=2.7799  train=0.0636  test=0.0614  best=0.0630  (17s, 17459 e
x/s)
    epoch   7/40  loss=2.7788  train=0.0644  test=0.0620  best=0.0630  (17s, 17490 e
x/s)
    epoch   8/40  loss=2.7774  train=0.0656  test=0.0631  best=0.0630  (17s, 17600 e
x/s)
    epoch   9/40  loss=2.7765  train=0.0656  test=0.0644  best=0.0630  (17s, 17621 e
x/s)
    epoch  10/40  loss=2.7756  train=0.0661  test=0.0633  best=0.0644  (17s, 17702 e
x/s)
    epoch  11/40  loss=2.7747  train=0.0672  test=0.0618  best=0.0644  (17s, 17608 e
x/s)
    epoch  12/40  loss=2.7738  train=0.0680  test=0.0626  best=0.0644  (17s, 17456 e
x/s)
    epoch  13/40  loss=2.7731  train=0.0687  test=0.0620  best=0.0644  (17s, 17454 e
x/s)
    epoch  14/40  loss=2.7723  train=0.0697  test=0.0620  best=0.0644  (17s, 17460 e
x/s)
    epoch  15/40  loss=2.7718  train=0.0698  test=0.0633  best=0.0644  (17s, 17448 e
x/s)
    epoch  16/40  loss=2.7710  train=0.0712  test=0.0626  best=0.0644  (17s, 17577 e
x/s)
    epoch  17/40  loss=2.7703  train=0.0719  test=0.0625  best=0.0644  (17s, 17584 e
x/s)
    epoch  18/40  loss=2.7698  train=0.0720  test=0.0629  best=0.0644  (17s, 17597 e
x/s)
    epoch  19/40  loss=2.7692  train=0.0723  test=0.0628  best=0.0644  (17s, 17614 e
x/s)
    epoch  20/40  loss=2.7685  train=0.0739  test=0.0636  best=0.0644  (17s, 17574 e
x/s)
    epoch  21/40  loss=2.7680  train=0.0735  test=0.0639  best=0.0644  (17s, 17594 e
x/s)
```

```
    epoch  22/40  loss=2.7674  train=0.0746  test=0.0640  best=0.0644  (17s, 17568 e
x/s)
    epoch  23/40  loss=2.7666  train=0.0760  test=0.0634  best=0.0644  (17s, 17607 e
x/s)
    epoch  24/40  loss=2.7657  train=0.0767  test=0.0626  best=0.0644  (17s, 17571 e
x/s)
    epoch  25/40  loss=2.7649  train=0.0777  test=0.0632  best=0.0644  (17s, 17576 e
x/s)
    epoch  26/40  loss=2.7638  train=0.0781  test=0.0646  best=0.0644  (17s, 17585 e
x/s)
    epoch  27/40  loss=2.7630  train=0.0792  test=0.0612  best=0.0644  (17s, 17597 e
x/s)
    epoch  28/40  loss=2.7621  train=0.0806  test=0.0621  best=0.0644  (17s, 17599 e
x/s)
    epoch  29/40  loss=2.7612  train=0.0805  test=0.0625  best=0.0644  (17s, 17548 e
x/s)
    epoch  30/40  loss=2.7600  train=0.0821  test=0.0625  best=0.0644  (17s, 17568 e
x/s)
    epoch  31/40  loss=2.7588  train=0.0830  test=0.0641  best=0.0644  (17s, 17614 e
x/s)
    epoch  32/40  loss=2.7563  train=0.0840  test=0.0621  best=0.0644  (17s, 17585 e
x/s)
    epoch  33/40  loss=2.7532  train=0.0849  test=0.0632  best=0.0644  (17s, 17607 e
x/s)
    epoch  34/40  loss=2.7380  train=0.0870  test=0.0703  best=0.0644  (17s, 17568 e
x/s)
    epoch  35/40  loss=2.5126  train=0.1080  test=0.1137  best=0.0703  (17s, 17574 e
x/s)
    epoch  36/40  loss=2.1532  train=0.1331  test=0.1234  best=0.1137  (17s, 17437 e
x/s)
    epoch  37/40  loss=2.0914  train=0.1381  test=0.1263  best=0.1234  (17s, 17572 e
x/s)
    epoch  38/40  loss=2.0815  train=0.1424  test=0.1279  best=0.1263  (17s, 17495 e
x/s)
    epoch  39/40  loss=2.0801  train=0.1450  test=0.1271  best=0.1279  (17s, 17597 e
x/s)
    epoch  40/40  loss=2.0754  train=0.1484  test=0.1245  best=0.1279  (17s, 17587 e
x/s)
  [B27_p4] Done. Final acc = 0.1279, best acc = 0.1279  [NEEDS RETRY]

======================================================================
Experiment: base=27, p=6 (n mod 64), v₂(B)=0
======================================================================
  [B27_p6] Existing run below threshold (score=0.0164); resuming / extending trainin
g.
  [B27_p6] Resumed from epoch 40
  [B27_p6] classes=64 | max_len=9 | params=12,663,360 | chance=0.0156
  [B27_p6] Done. Final acc = 0.0164, best acc = 0.0168  [NEEDS RETRY]

======================================================================
Experiment: base=27, p=8 (n mod 256), v₂(B)=0
======================================================================
  [B27_p8] Existing run below threshold (score=0.0051); resuming / extending trainin
g.
  [B27_p8] Resumed from epoch 40
  [B27_p8] classes=256 | max_len=9 | params=12,761,856 | chance=0.0039
```

```
  [B27_p8] Done. Final acc = 0.0051, best acc = 0.0051  [NEEDS RETRY]

Sweep finished in 0.36 hours
Successful runs now: 21 / 35
```

8. Load Results (For Post-Disconnect Analysis)

If you got disconnected and are restarting, run this cell to load all saved results.

In [11]:
```
# Reload all results from saved JSON files
all_results = {}
for base in BASES:
    for p in P_VALUES:
        result_path = os.path.join(SAVE_DIR, f'B{base}_p{p}_results.json')
        if os.path.exists(result_path):
            with open(result_path, 'r') as f:
                r = json.load(f)
            all_results[(base, p)] = r

print(f'Loaded {len(all_results)} / {len(BASES)*len(P_VALUES)} experiment results')

successful = []
unsuccessful = []

for key, r in sorted(all_results.items()):
    score = get_result_score(r)
    if experiment_succeeded(r, SUCCESS_ACC_THRESHOLD):
        successful.append((key, score))
    else:
        unsuccessful.append((key, score))

print(f'Successful (score >= {SUCCESS_ACC_THRESHOLD:.2f}): {len(successful)}')
print(f'Need retrain / unfinished: {len(unsuccessful)}')

if unsuccessful:
    print('\nRetry list:')
    print([key for key, _ in unsuccessful])

print('\nQuick summary')
for (b, p), r in sorted(all_results.items()):
    final_acc = r.get('final_acc', None)
    best_acc = r.get('best_acc', None)
    score = get_result_score(r)
    status = 'OK' if experiment_succeeded(r, SUCCESS_ACC_THRESHOLD) else 'RETRY'
    final_str = f'{final_acc:.4f}' if isinstance(final_acc, float) else str(final_a
    best_str = f'{best_acc:.4f}' if isinstance(best_acc, float) else str(best_acc)
    print(f'  B={b:>2}, p={p}: final={final_str:>7}  best={best_str:>7}  score={sco
```

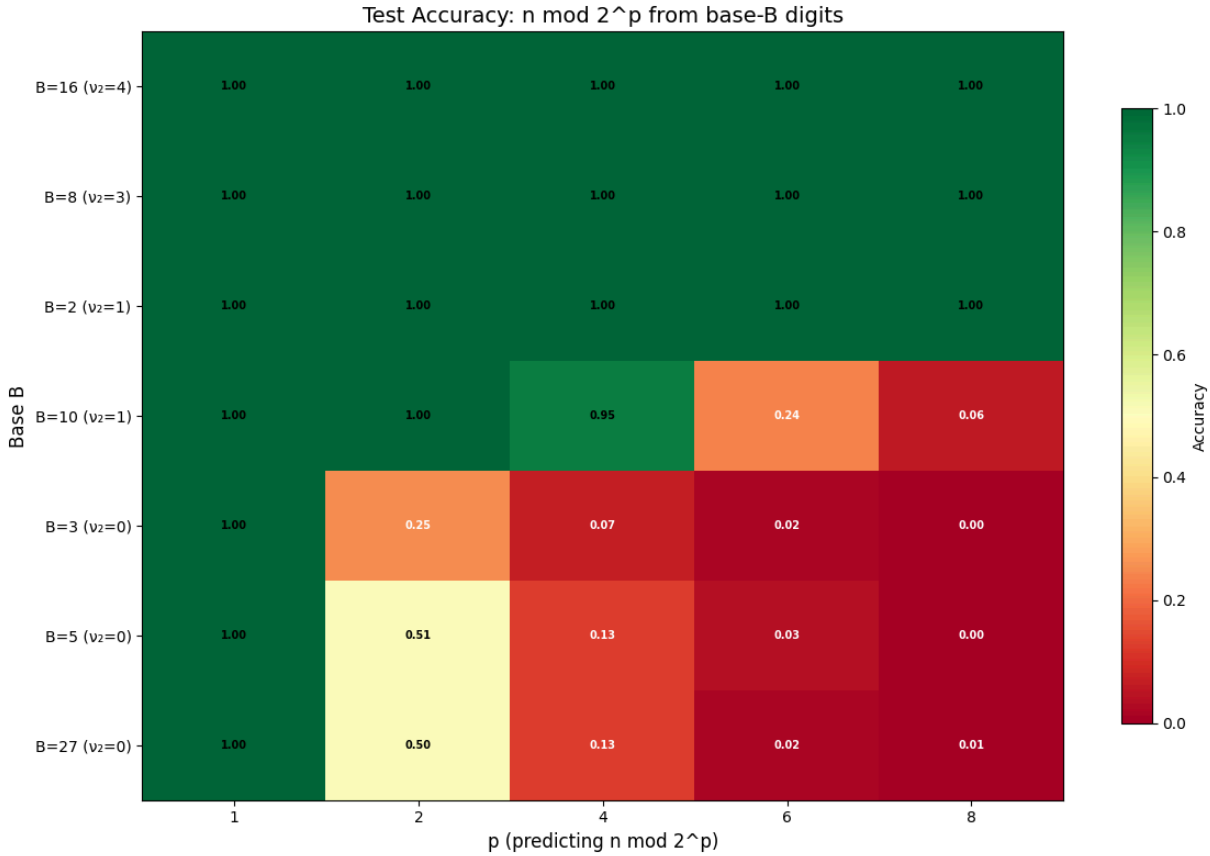
```
Loaded 35 / 35 experiment results
Successful (score >= 0.90): 21
Need retrain / unfinished: 14

Retry list:
[(3, 2), (3, 4), (3, 6), (3, 8), (5, 2), (5, 4), (5, 6), (5, 8), (10, 6), (10, 8),
(27, 2), (27, 4), (27, 6), (27, 8)]

Quick summary
  B= 2, p=1: final= 0.9998  best= 0.9998  score= 0.9998  OK
  B= 2, p=2: final= 0.9990  best= 0.9990  score= 0.9990  OK
  B= 2, p=4: final= 0.9988  best= 0.9988  score= 0.9988  OK
  B= 2, p=6: final= 0.9976  best= 0.9976  score= 0.9976  OK
  B= 2, p=8: final= 0.9978  best= 0.9978  score= 0.9978  OK
  B= 3, p=1: final= 0.9989  best= 0.9989  score= 0.9989  OK
  B= 3, p=2: final= 0.2544  best= 0.2548  score= 0.2548  RETRY
  B= 3, p=4: final= 0.0686  best= 0.0686  score= 0.0686  RETRY
  B= 3, p=6: final= 0.0160  best= 0.0170  score= 0.0170  RETRY
  B= 3, p=8: final= 0.0034  best= 0.0043  score= 0.0043  RETRY
  B= 5, p=1: final= 0.9993  best= 0.9993  score= 0.9993  OK
  B= 5, p=2: final= 0.5060  best= 0.5060  score= 0.5060  RETRY
  B= 5, p=4: final= 0.1270  best= 0.1277  score= 0.1277  RETRY
  B= 5, p=6: final= 0.0323  best= 0.0323  score= 0.0323  RETRY
  B= 5, p=8: final= 0.0041  best= 0.0041  score= 0.0041  RETRY
  B= 8, p=1: final= 0.9992  best= 0.9992  score= 0.9992  OK
  B= 8, p=2: final= 0.9995  best= 0.9995  score= 0.9995  OK
  B= 8, p=4: final= 0.9996  best= 0.9996  score= 0.9996  OK
  B= 8, p=6: final= 0.9997  best= 0.9997  score= 0.9997  OK
  B= 8, p=8: final= 0.9996  best= 0.9996  score= 0.9996  OK
  B=10, p=1: final= 0.9997  best= 0.9997  score= 0.9997  OK
  B=10, p=2: final= 0.9988  best= 0.9988  score= 0.9988  OK
  B=10, p=4: final= 0.9467  best= 0.9467  score= 0.9467  OK
  B=10, p=6: final= 0.2401  best= 0.2404  score= 0.2404  RETRY
  B=10, p=8: final= 0.0607  best= 0.0615  score= 0.0615  RETRY
  B=16, p=1: final= 0.9994  best= 0.9994  score= 0.9994  OK
  B=16, p=2: final= 0.9994  best= 0.9994  score= 0.9994  OK
  B=16, p=4: final= 0.9997  best= 0.9997  score= 0.9997  OK
  B=16, p=6: final= 0.9994  best= 0.9994  score= 0.9994  OK
  B=16, p=8: final= 0.9990  best= 0.9990  score= 0.9990  OK
  B=27, p=1: final= 0.9991  best= 0.9991  score= 0.9991  OK
  B=27, p=2: final= 0.5046  best= 0.5046  score= 0.5046  RETRY
  B=27, p=4: final= 0.1279  best= 0.1279  score= 0.1279  RETRY
  B=27, p=6: final= 0.0164  best= 0.0168  score= 0.0168  RETRY
  B=27, p=8: final= 0.0051  best= 0.0051  score= 0.0051  RETRY
```

9. Visualization — Accuracy Heatmap (Central Figure)

In [12]:
```
def plot_accuracy_heatmap(results, bases, p_values):
    """Plot the B vs p accuracy heatmap — the central figure of the project."""
    grid = np.full((len(bases), len(p_values)), np.nan)

    for i, b in enumerate(bases):
        for j, p in enumerate(p_values):
```

```
            if (b, p) in results:
                acc = results[(b, p)].get('final_acc', np.nan)
                grid[i, j] = acc if isinstance(acc, (int, float)) else np.nan

    fig, ax = plt.subplots(figsize=(12, 8))
    im = ax.imshow(grid, cmap='RdYlGn', vmin=0, vmax=1, aspect='auto')

    ax.set_xticks(range(len(p_values)))
    ax.set_xticklabels([str(p) for p in p_values])
    ax.set_yticks(range(len(bases)))
    # Label bases with their 2-adic valuation
    ylabels = [f'B={b} (v₂={nu_2(b)})' for b in bases]
    ax.set_yticklabels(ylabels)
    ax.set_xlabel('p (predicting n mod 2^p)', fontsize=12)
    ax.set_ylabel('Base B', fontsize=12)
    ax.set_title('Test Accuracy: n mod 2^p from base-B digits', fontsize=14)

    # Annotate cells
    for i in range(len(bases)):
        for j in range(len(p_values)):
            val = grid[i, j]
            if not np.isnan(val):
                color = 'white' if val < 0.5 else 'black'
                ax.text(j, i, f'{val:.2f}', ha='center', va='center',
                        fontsize=7, color=color, fontweight='bold')

    plt.colorbar(im, ax=ax, label='Accuracy', shrink=0.8)
    plt.tight_layout()

    # Save
    fig_path = os.path.join(SAVE_DIR, 'accuracy_heatmap.png')
    plt.savefig(fig_path, dpi=150, bbox_inches='tight')
    print(f'Saved to {fig_path}')
    plt.show()


# Sort bases for display: powers of 2, then even by v₂, then odd
display_bases = sorted(BASES, key=lambda b: (-nu_2(b), b))
plot_accuracy_heatmap(all_results, display_bases, P_VALUES)
```

```
Saved to /content/drive/MyDrive/binary_suffix_experiment/accuracy_heatmap.png
```

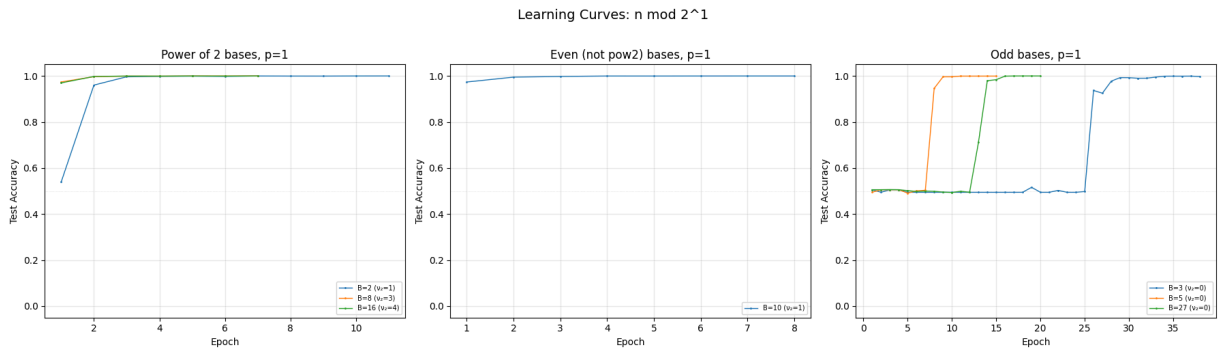


10. Visualization — Learning Curves

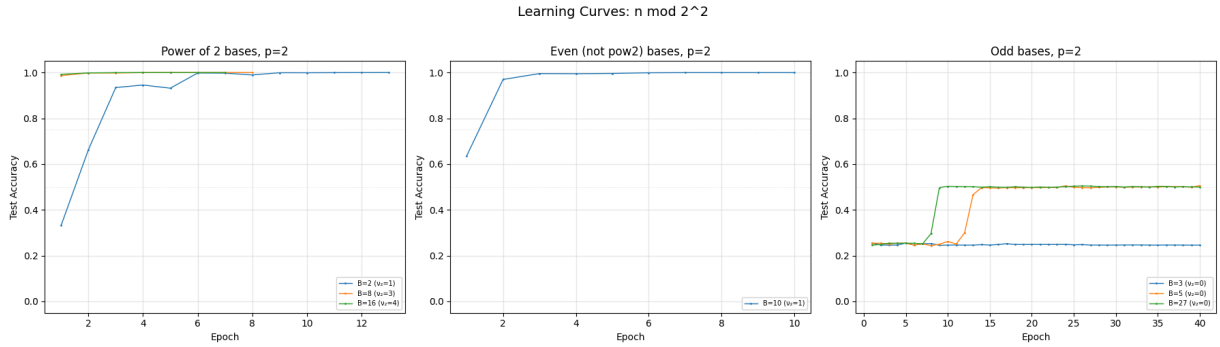
In [13]:
```
def plot_learning_curves(results, p_fixed=4):
    """Plot learning curves for a fixed p, grouped by v₂(B)."""
    fig, axes = plt.subplots(1, 3, figsize=(18, 5))

    groups = {
        'Power of 2': [b for b in BASES if b & (b-1) == 0],
        'Even (not pow2)': [b for b in BASES if b % 2 == 0 and b & (b-1) != 0],
        'Odd': [b for b in BASES if b % 2 != 0],
    }

    for ax, (group_name, bases) in zip(axes, groups.items()):
        for b in bases:
            key = (b, p_fixed)
            if key in results and 'history' in results[key]:
                h = results[key]['history']
                epochs = [e + 1 for e in h['epoch']]
                ax.plot(epochs, h['test_acc'], label=f'B={b} (v₂={nu_2(b)})',
                        marker='.', markersize=2, linewidth=1)

        # Add theoretical step levels for reference
        for j in range(1, p_fixed + 1):
            level = 1 - 1/(2**j)
            ax.axhline(y=level, color='gray', linestyle=':', alpha=0.3, linewidth=0

        ax.set_xlabel('Epoch')
        ax.set_ylabel('Test Accuracy')
```

```
        ax.set_title(f'{group_name} bases, p={p_fixed}')
        ax.legend(fontsize=7, loc='lower right')
        ax.set_ylim(-0.05, 1.05)
        ax.grid(True, alpha=0.3)

    plt.suptitle(f'Learning Curves: n mod 2^{p_fixed}', fontsize=14, y=1.02)
    plt.tight_layout()
    fig_path = os.path.join(SAVE_DIR, f'learning_curves_p{p_fixed}.png')
    plt.savefig(fig_path, dpi=150, bbox_inches='tight')
    print(f'Saved to {fig_path}')
    plt.show()


# Plot for several p values
for p_val in [1, 2, 4, 6]:
    if any((b, p_val) in all_results for b in BASES):
        plot_learning_curves(all_results, p_fixed=p_val)
```

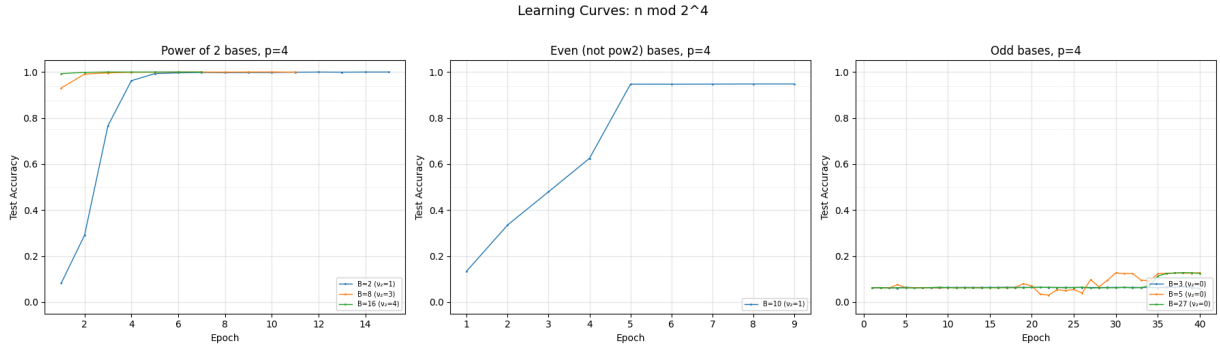
```
Saved to /content/drive/MyDrive/binary_suffix_experiment/learning_curves_p1.png
```



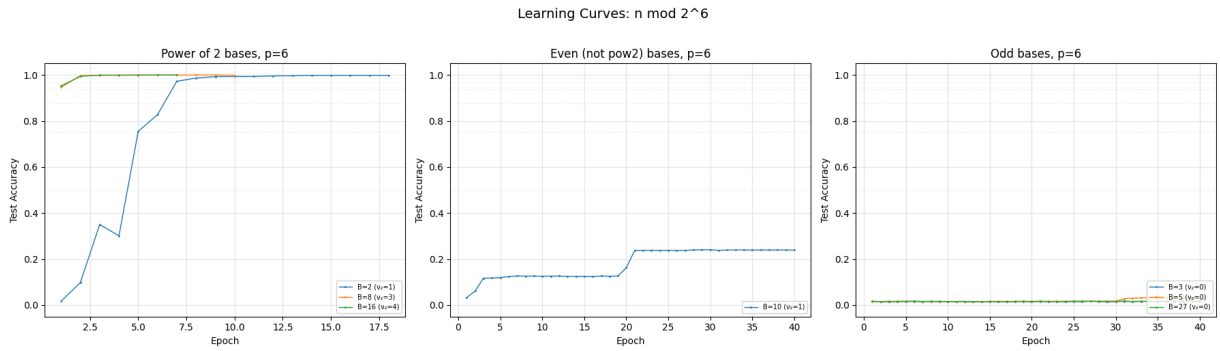

```
Saved to /content/drive/MyDrive/binary_suffix_experiment/learning_curves_p2.png
```




```
Saved to /content/drive/MyDrive/binary_suffix_experiment/learning_curves_p4.png
```



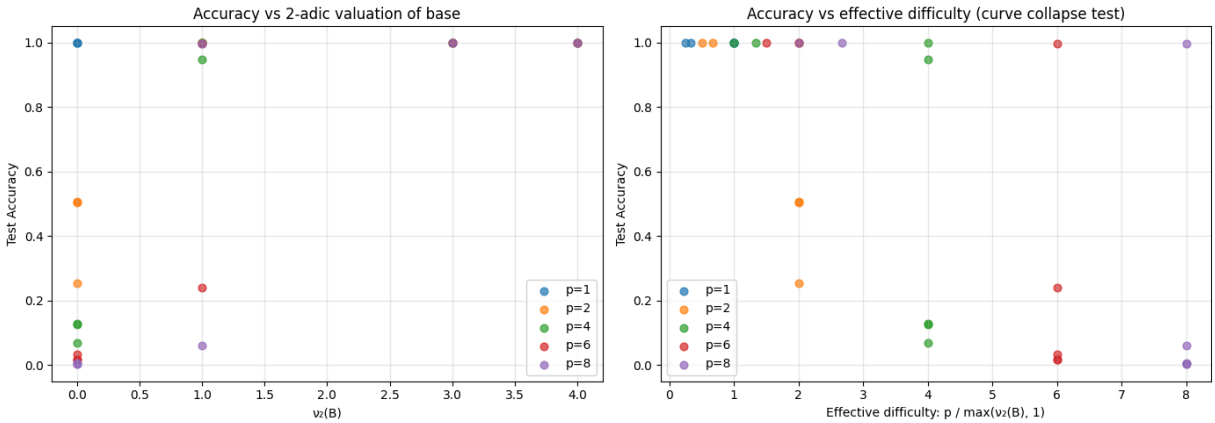

```
Saved to /content/drive/MyDrive/binary_suffix_experiment/learning_curves_p6.png
```



11. Visualization — Accuracy vs $\nu_2(B)$ (Testing Primary Hypothesis)

In [14]:
```
def plot_accuracy_vs_nu2(results, p_values_to_plot=[1, 2, 4, 6, 8]):
    """Test whether accuracy is primarily determined by v₂(B)."""
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))

    # Left: accuracy vs v₂(B) for different p
    ax = axes[0]
    for p in p_values_to_plot:
        nu2_vals, accs = [], []
        for b in BASES:
            if (b, p) in results:
                acc = results[(b, p)].get('final_acc', None)
                if acc is not None and isinstance(acc, (int, float)):
                    nu2_vals.append(nu_2(b))
                    accs.append(acc)
        if nu2_vals:
            ax.scatter(nu2_vals, accs, label=f'p={p}', s=40, alpha=0.7)
    ax.set_xlabel('v₂(B)')
    ax.set_ylabel('Test Accuracy')
    ax.set_title('Accuracy vs 2-adic valuation of base')
    ax.legend()
    ax.grid(True, alpha=0.3)
    ax.set_ylim(-0.05, 1.05)

    # Right: accuracy vs effective difficulty p / max(v₂(B), 1)
    ax = axes[1]
    for p in p_values_to_plot:
        diffs, accs = [], []
        for b in BASES:
            if (b, p) in results:
                acc = results[(b, p)].get('final_acc', None)
                if acc is not None and isinstance(acc, (int, float)):
                    eff_diff = p / max(nu_2(b), 1)
                    diffs.append(eff_diff)
                    accs.append(acc)
        if diffs:
            ax.scatter(diffs, accs, label=f'p={p}', s=40, alpha=0.7)
    ax.set_xlabel('Effective difficulty: p / max(v₂(B), 1)')
    ax.set_ylabel('Test Accuracy')
    ax.set_title('Accuracy vs effective difficulty (curve collapse test)')
    ax.legend()
```

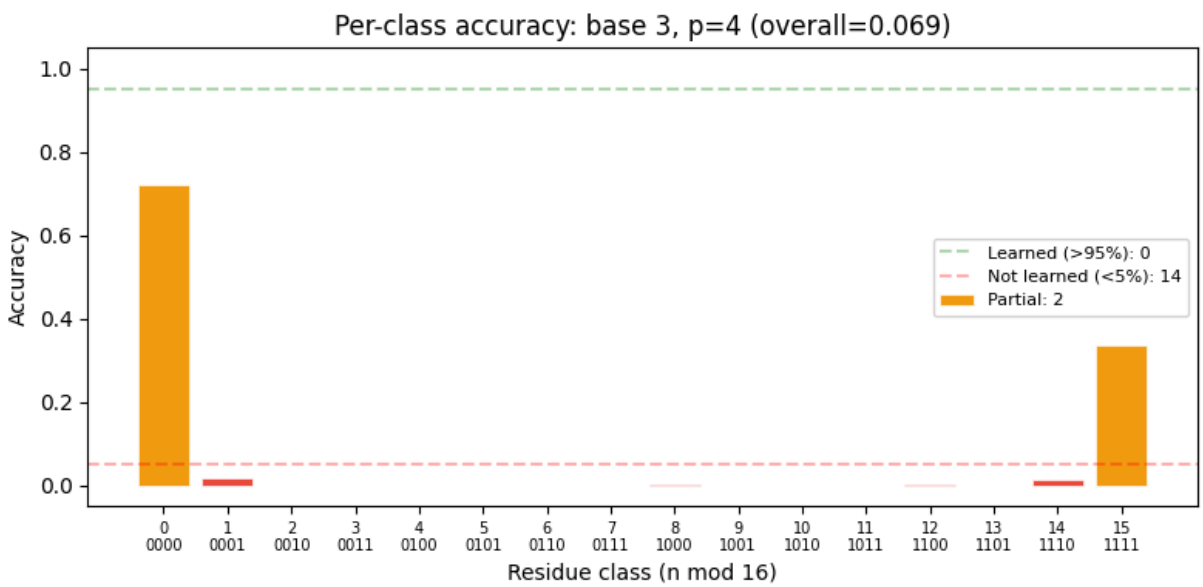
```
    ax.grid(True, alpha=0.3)
    ax.set_ylim(-0.05, 1.05)

    plt.tight_layout()
    fig_path = os.path.join(SAVE_DIR, 'accuracy_vs_nu2.png')
    plt.savefig(fig_path, dpi=150, bbox_inches='tight')
    plt.show()


plot_accuracy_vs_nu2(all_results)
```

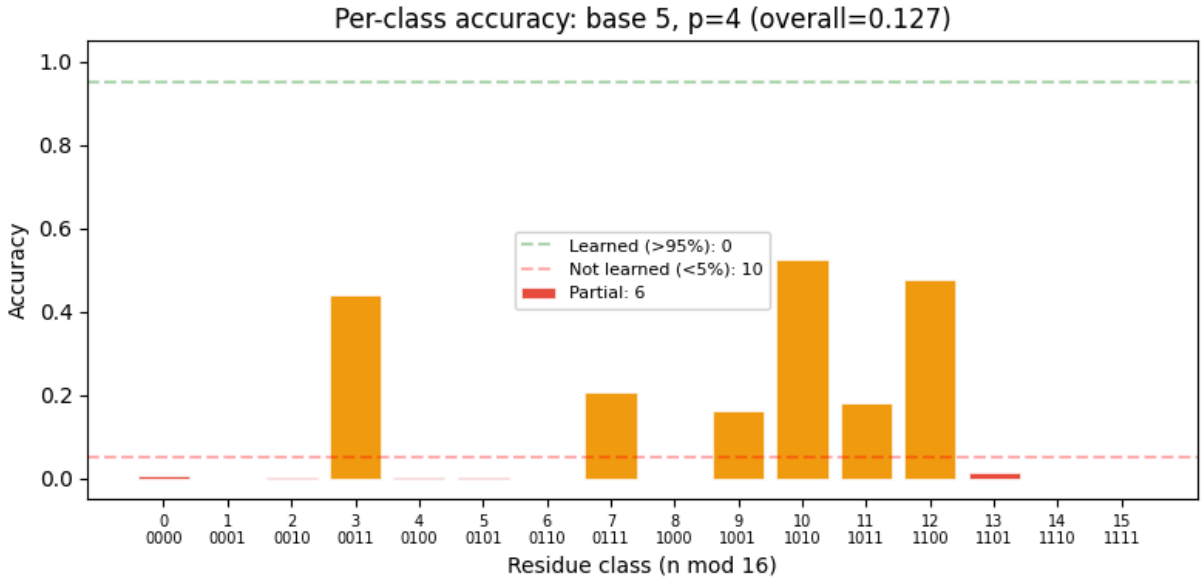


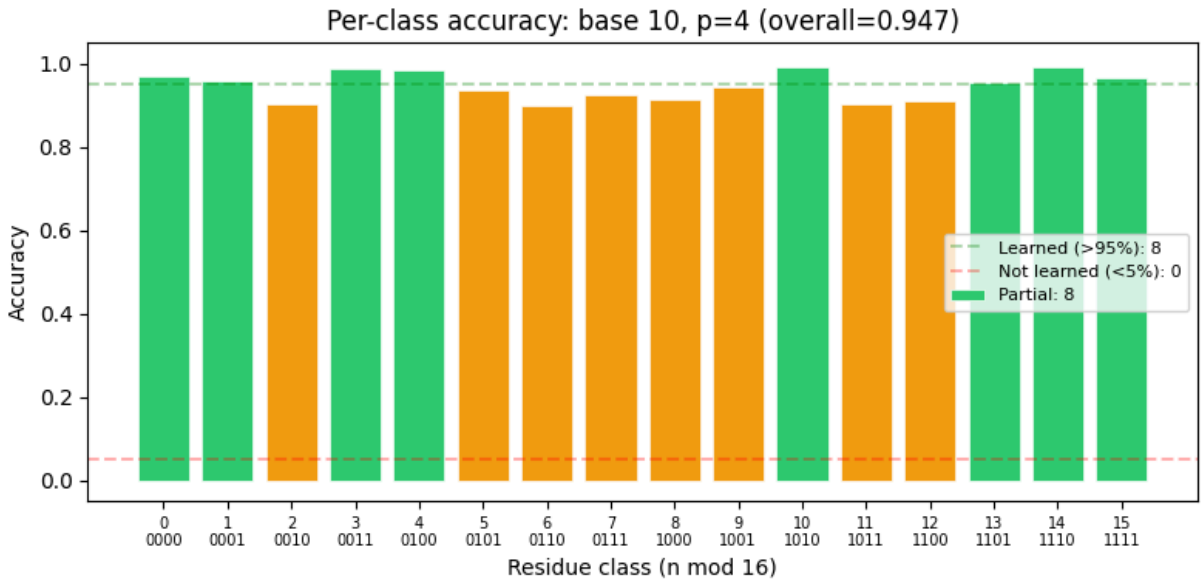
12. Per-Class Accuracy Analysis (Step-Like Learning Patterns)

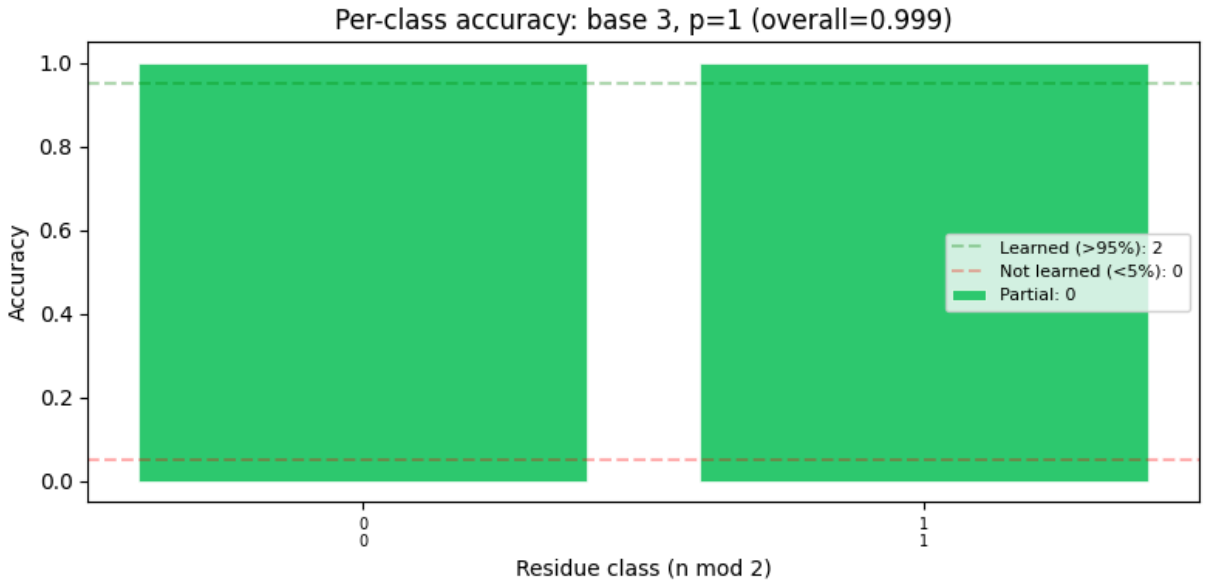
In [15]:
```
def plot_per_class_accuracy(results, base, p):
    """Show per-class accuracy for a specific (base, p) experiment."""
    key = (base, p)
    if key not in results:
        print(f'No results for B={base}, p={p}')
        return

    r = results[key]
    class_acc = r.get('final_class_acc', {})
    if not class_acc:
        print(f'No per-class data for B={base}, p={p}')
        return

    num_classes = 2 ** p
    classes = list(range(num_classes))
    accs = [class_acc.get(str(c), 0) for c in classes]

    # Color by binary suffix pattern
    fig, ax = plt.subplots(figsize=(max(8, num_classes * 0.3), 4))
    colors = ['#2ecc71' if a > 0.95 else '#e74c3c' if a < 0.05 else '#f39c12' for a
    bars = ax.bar(classes, accs, color=colors, edgecolor='white', linewidth=0.5)

    ax.set_xlabel(f'Residue class (n mod {num_classes})')
    ax.set_ylabel('Accuracy')
    ax.set_title(f'Per-class accuracy: base {base}, p={p} (overall={r["final_acc"]:
    ax.set_ylim(-0.05, 1.05)
    ax.axhline(y=0.95, color='green', linestyle='--', alpha=0.3, label='95% thresho
```

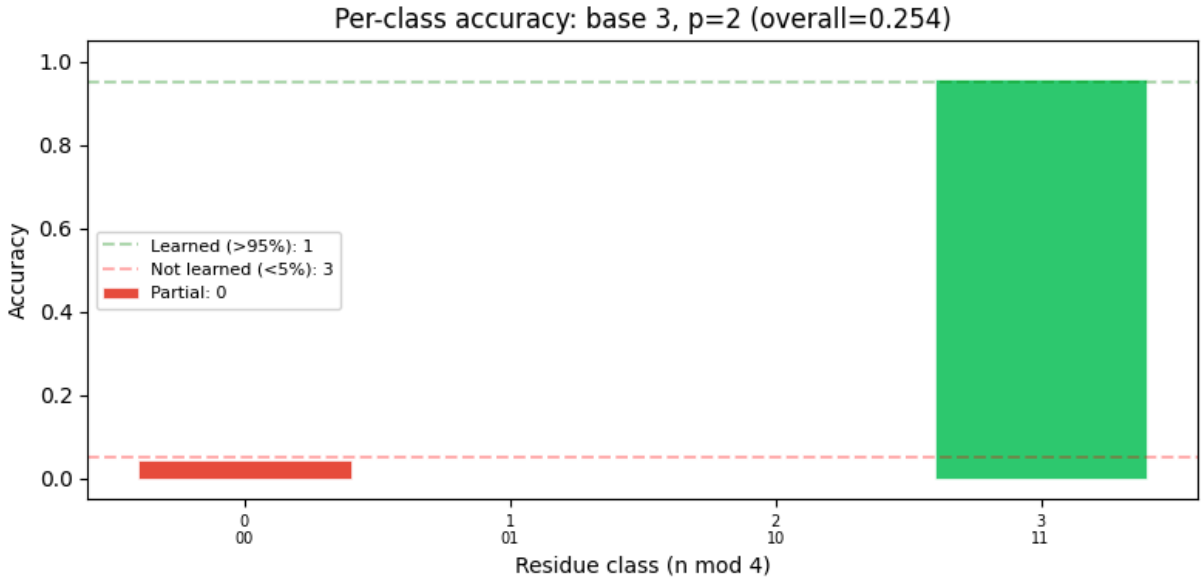
```
    ax.axhline(y=0.05, color='red', linestyle='--', alpha=0.3, label='5% threshold'

    # Count learned vs not-learned classes
    learned = sum(1 for a in accs if a > 0.95)
    not_learned = sum(1 for a in accs if a < 0.05)
    partial = num_classes - learned - not_learned
    ax.legend([f'Learned (>95%): {learned}', f'Not learned (<5%): {not_learned}',
              f'Partial: {partial}'], fontsize=8)

    if num_classes <= 32:
        ax.set_xticks(classes)
        # Show binary representation on x-axis for small num_classes
        if num_classes <= 16:
            ax.set_xticklabels([f'{c}\n{c:0{p}b}' for c in classes], fontsize=7)

    plt.tight_layout()
    fig_path = os.path.join(SAVE_DIR, f'per_class_B{base}_p{p}.png')
    plt.savefig(fig_path, dpi=150, bbox_inches='tight')
    plt.show()


# Plot for interesting cases
interesting_cases = [
    (3, 4), (5, 4), (7, 4),      # odd bases
    (10, 4), (6, 4), (12, 4),    # even non-power-of-2
    (3, 1), (3, 2), (3, 6),      # base 3 at different p
    (27, 4), (9, 4),             # odd bases, higher
]
for b, p in interesting_cases:
    if (b, p) in all_results:
        plot_per_class_accuracy(all_results, b, p)
```

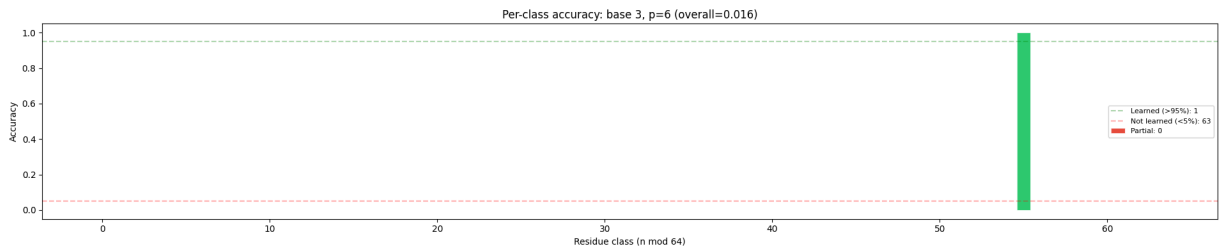


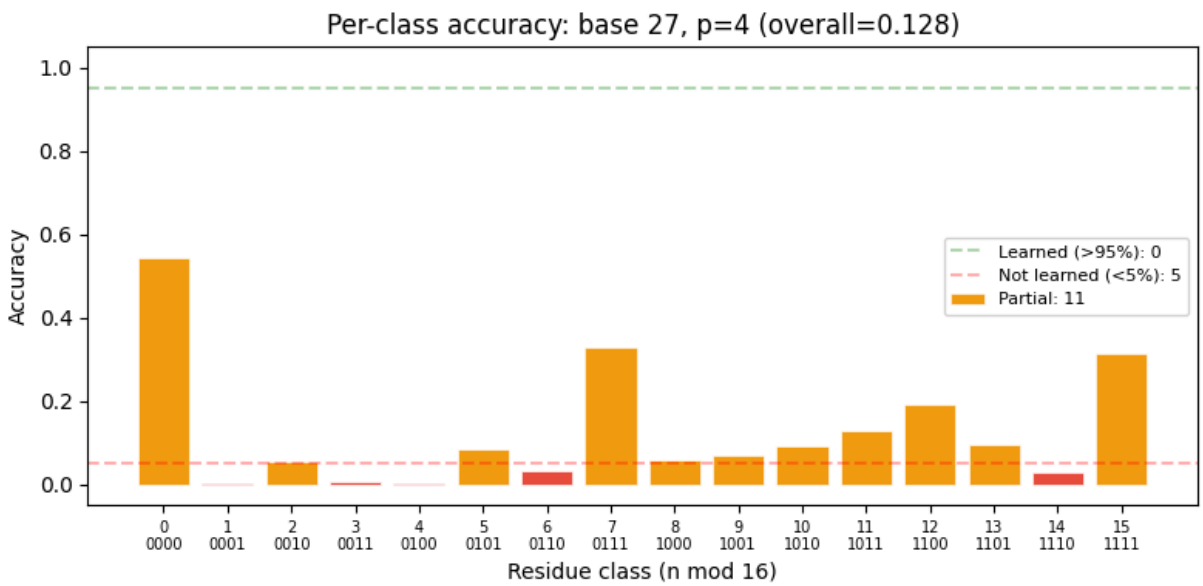

Per-class accuracy: base 5, p=4 (overall=0.127)
Accuracy
Learned (>95%): 0
Not learned (<5%): 10
Partial: 6
0 0000
1 0001
2 0010
3 0011
4 0100
5 0101
6 0110
7 0111
8 1000
9 1001
10 1010
11 1011
12 1100
13 1101
14 1110
15 1111
Residue class (n mod 16)


Per-class accuracy: base 10, p=4 (overall=0.947)
Accuracy
Learned (>95%): 8
Not learned (<5%): 0
Partial: 8
0 0000
1 0001
2 0010
3 0011
4 0100
5 0101
6 0110
7 0111
8 1000
9 1001
10 1010
11 1011
12 1100
13 1101
14 1110
15 1111
Residue class (n mod 16)


Per-class accuracy: base 3, p=1 (overall=0.999)
Accuracy
Learned (>95%): 2
Not learned (<5%): 0
Partial: 0
0 0
1 1
Residue class (n mod 2)







13. Linear Probing of Hidden Representations

Train logistic regression probes on frozen encoder hidden states to predict individual bits of $n \bmod 2^p$. This tests whether the model internally constructs a binary representation, even when inputs are in a non-binary base.

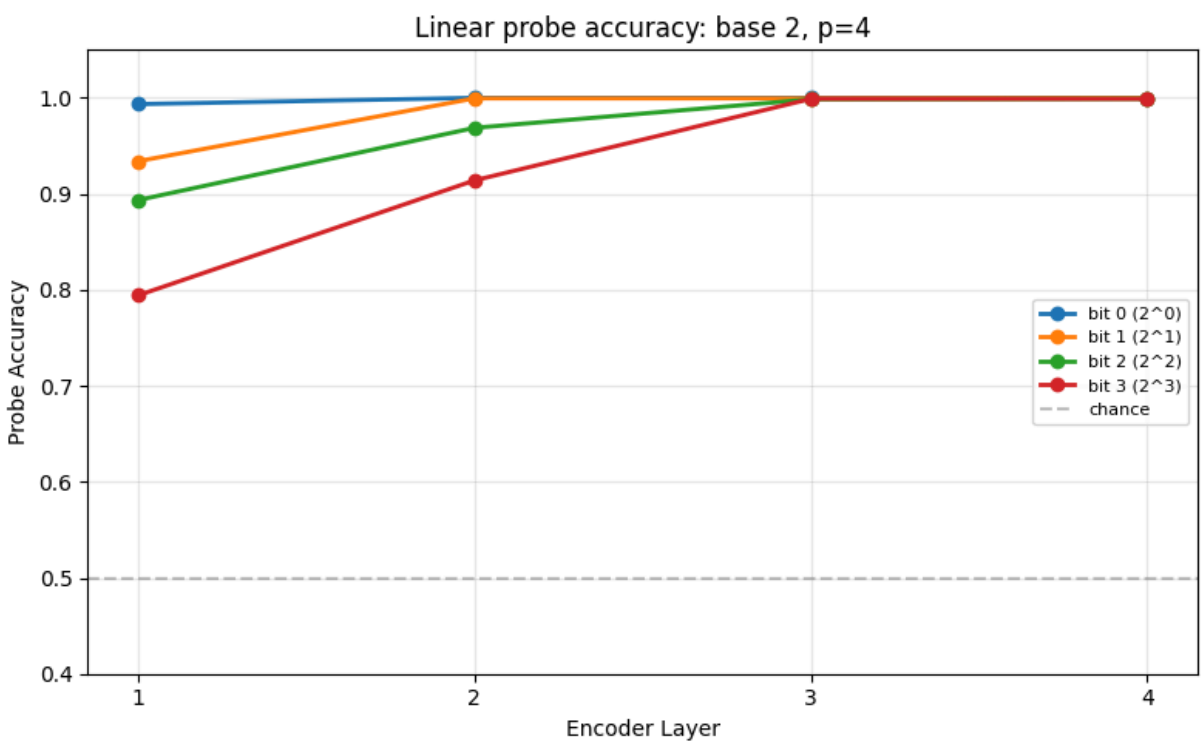
In [16]:
```
def run_linear_probes(base, p, save_dir=SAVE_DIR, probe_size=20000):
    """
    Train linear probes on each encoder layer to predict individual bits.
```

```
    Returns: dict mapping (layer_idx, bit_idx) -> probe accuracy
    """
    exp_name = f'B{base}_p{p}'
    model_path = os.path.join(save_dir, f'{exp_name}_model.pt')

    if not os.path.exists(model_path):
        print(f'  No saved model for {exp_name}')
        return None

    num_classes = 2 ** p
    msl = max_seq_len(N_MAX, base) + 2
    vocab_size = base + 3
    pad_id = base + 2

    # Load trained model
    model = BinarySuffixTransformer(
        vocab_size=vocab_size, num_classes=num_classes,
        d_model=D_MODEL, nhead=N_HEADS, num_layers=ENC_LAYERS,
        dim_feedforward=DIM_FF, max_len=msl, dropout=0.0, pad_id=pad_id,
    ).to(device)
    model.load_state_dict(torch.load(model_path, map_location=device, weights_only=
    model.eval()

    # Generate probe data
    X_probe, y_probe, ns_probe = make_eval_data(base, p, N_MAX, probe_size, msl, se

    # Extract bit labels for each bit position
    bit_labels = {}  # bit_idx -> tensor of 0/1
    for bit_idx in range(p):
        bit_labels[bit_idx] = ((ns_probe >> bit_idx) & 1).long()

    # Extract representations from each layer
    all_reps = [[] for _ in range(ENC_LAYERS)]
    batch_size = 1024

    with torch.no_grad():
        for i in range(0, len(X_probe), batch_size):
            x = X_probe[i:i+batch_size].to(device)
            layer_reps = model.get_layer_representations(x)
            for l_idx, rep in enumerate(layer_reps):
                all_reps[l_idx].append(rep.cpu())

    for l_idx in range(ENC_LAYERS):
        all_reps[l_idx] = torch.cat(all_reps[l_idx], dim=0)

    # Train linear probes
    probe_results = {}
    split = int(0.8 * probe_size)

    for l_idx in range(ENC_LAYERS):
        for bit_idx in range(p):
            X_train = all_reps[l_idx][:split]
            y_train = bit_labels[bit_idx][:split]
            X_test = all_reps[l_idx][split:]
            y_test = bit_labels[bit_idx][split:]
```

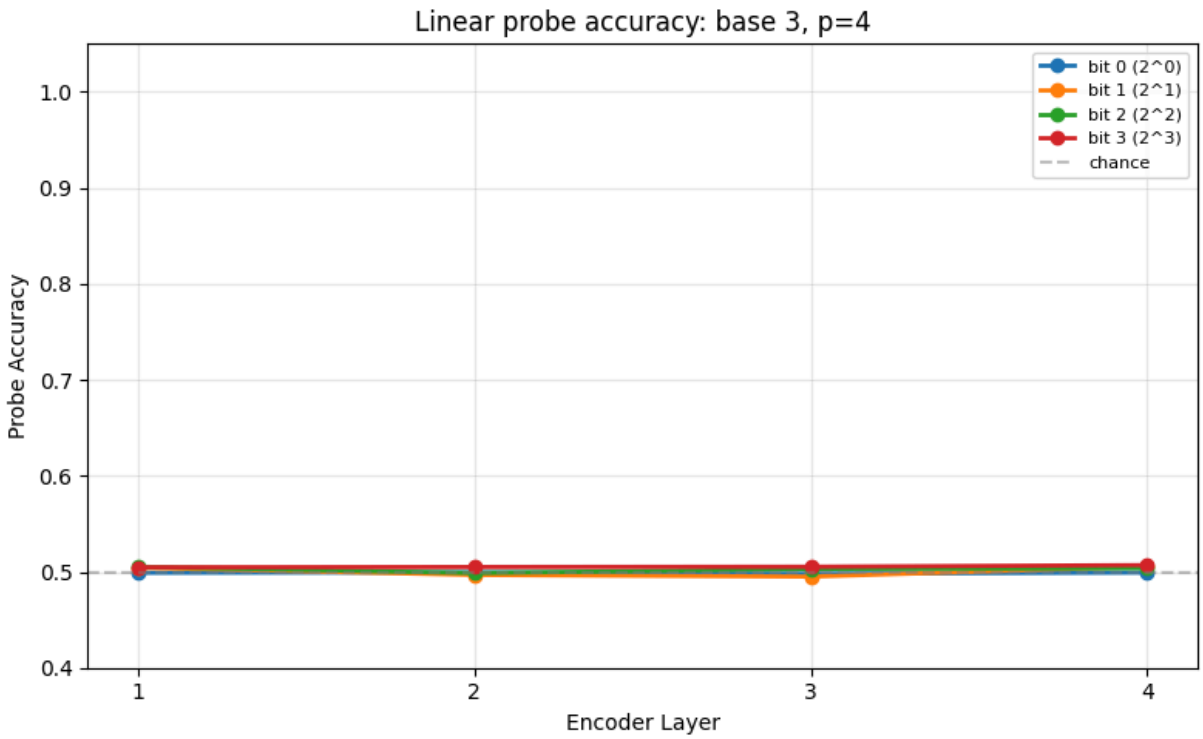
```
            # Simple logistic regression
            probe = nn.Linear(D_MODEL, 2).to(device)
            probe_opt = torch.optim.Adam(probe.parameters(), lr=1e-3)
            probe_criterion = nn.CrossEntropyLoss()

            # Train for a few epochs
            for _ in range(20):
                for j in range(0, len(X_train), 512):
                    xb = X_train[j:j+512].to(device)
                    yb = y_train[j:j+512].to(device)
                    probe_opt.zero_grad()
                    loss = probe_criterion(probe(xb), yb)
                    loss.backward()
                    probe_opt.step()

            # Evaluate probe
            with torch.no_grad():
                preds = probe(X_test.to(device)).argmax(-1)
                acc = (preds == y_test.to(device)).float().mean().item()

            probe_results[(l_idx, bit_idx)] = acc
            del probe

    del model
    torch.cuda.empty_cache()

    return probe_results


def plot_probe_results(probe_results, base, p):
    """Plot probe accuracy by layer and bit position."""
    if probe_results is None:
        return

    fig, ax = plt.subplots(figsize=(8, 5))

    for bit_idx in range(p):
        layers = list(range(ENC_LAYERS))
        accs = [probe_results.get((l, bit_idx), 0.5) for l in layers]
        ax.plot([l+1 for l in layers], accs, marker='o',
                label=f'bit {bit_idx} (2^{bit_idx})', linewidth=2)

    ax.axhline(y=0.5, color='gray', linestyle='--', alpha=0.5, label='chance')
    ax.set_xlabel('Encoder Layer')
    ax.set_ylabel('Probe Accuracy')
    ax.set_title(f'Linear probe accuracy: base {base}, p={p}')
    ax.legend(fontsize=8)
    ax.set_ylim(0.4, 1.05)
    ax.set_xticks(range(1, ENC_LAYERS + 1))
    ax.grid(True, alpha=0.3)

    plt.tight_layout()
    fig_path = os.path.join(SAVE_DIR, f'probes_B{base}_p{p}.png')
    plt.savefig(fig_path, dpi=150, bbox_inches='tight')
    plt.show()
```

In [17]:
```
if RUN_PROBES:
    # Run probes on a selection of interesting experiments
    probe_cases = [
        (2, 4),   # trivial baseline: should show perfect probes from layer 1
        (3, 4),   # hardest: odd base, p=4
        (5, 4),   # another odd base
        (10, 4),  # even, not power of 2
        (8, 4),   # power of 2
        (3, 2),   # odd base, easier p
        (27, 4),  # large odd base
    ]

    all_probe_results = {}
    for base, p in probe_cases:
        print(f'\nProbing B={base}, p={p}...')
        pr = run_linear_probes(base, p)
        if pr is not None:
            all_probe_results[(base, p)] = pr
            plot_probe_results(pr, base, p)

            # Print summary
            for bit_idx in range(p):
                layer_accs = [pr.get((l, bit_idx), 0.5) for l in range(ENC_LAYERS)]
                print(f'  bit {bit_idx}: layers = {[f"{a:.3f}" for a in layer_accs]
else:
    print('Probe analysis skipped (RUN_PROBES=False)')
```

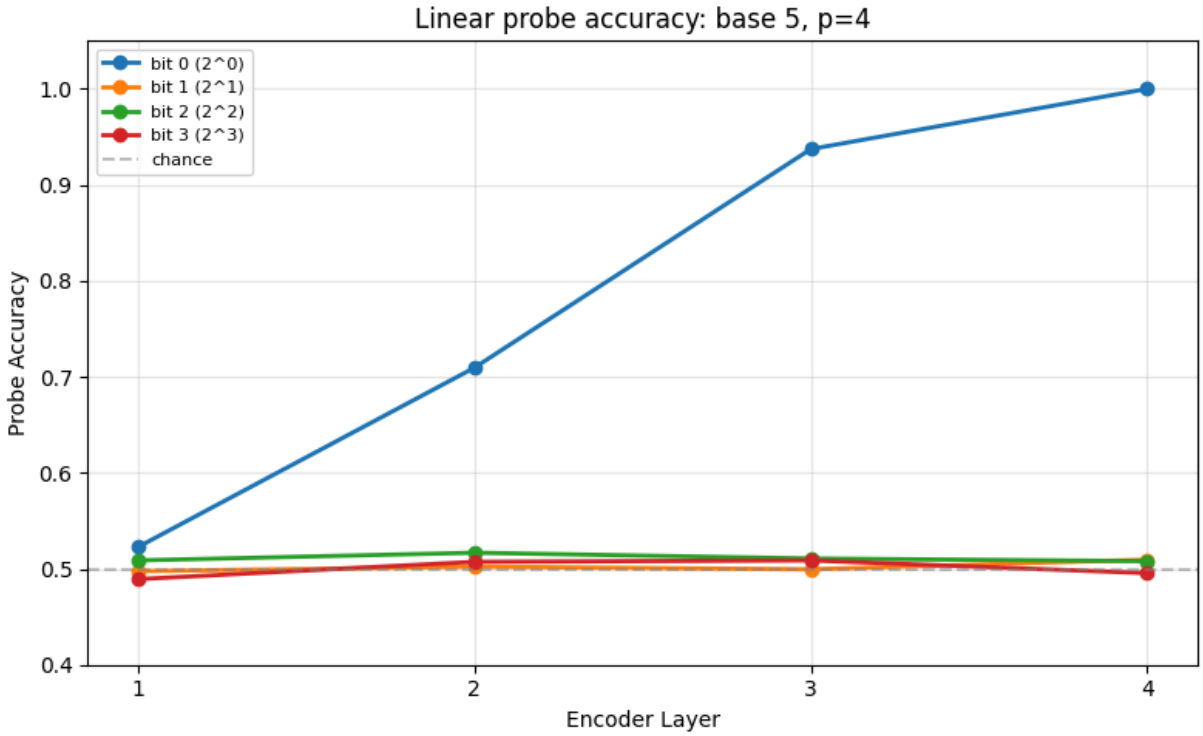
```
Probing B=2, p=4...
```



```
  bit 0: layers = ['0.994', '1.000', '1.000', '1.000']
  bit 1: layers = ['0.934', '1.000', '1.000', '1.000']
  bit 2: layers = ['0.894', '0.969', '0.999', '0.999']
  bit 3: layers = ['0.794', '0.914', '0.999', '0.999']

Probing B=3, p=4...
```

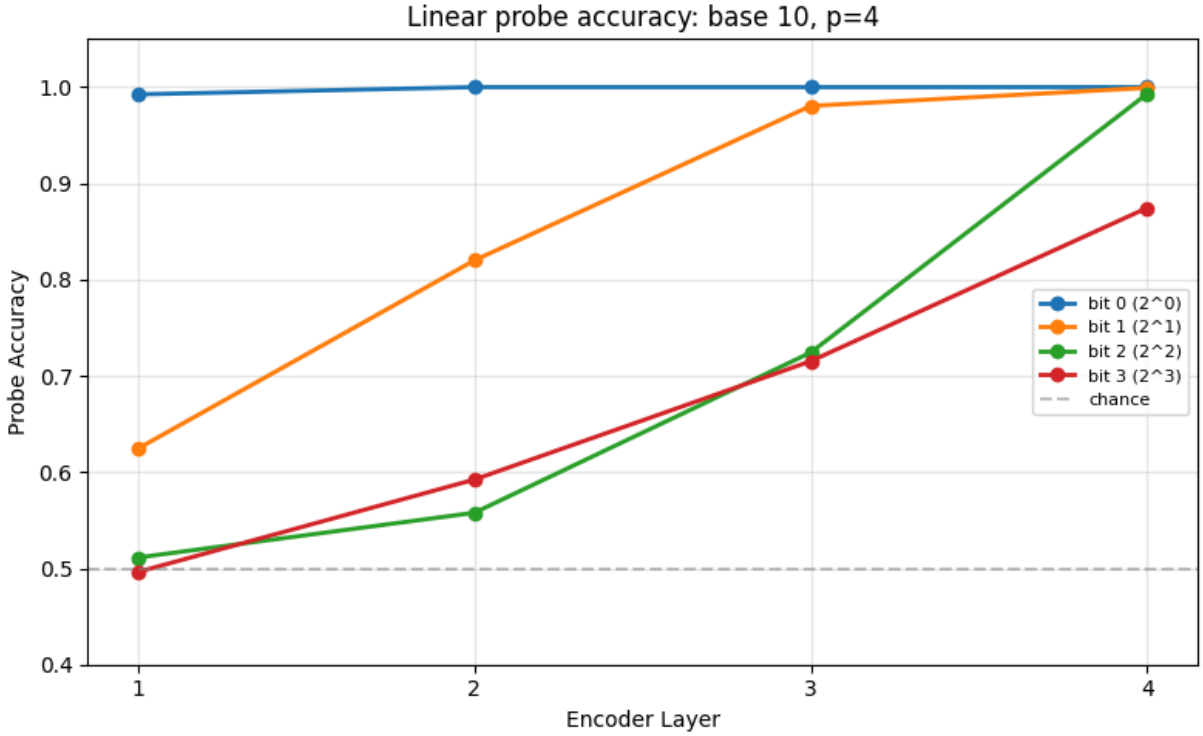



```
  bit 0: layers = ['0.499', '0.500', '0.498', '0.499']
  bit 1: layers = ['0.504', '0.497', '0.495', '0.506']
  bit 2: layers = ['0.505', '0.499', '0.503', '0.504']
  bit 3: layers = ['0.505', '0.505', '0.505', '0.507']

Probing B=5, p=4...
```

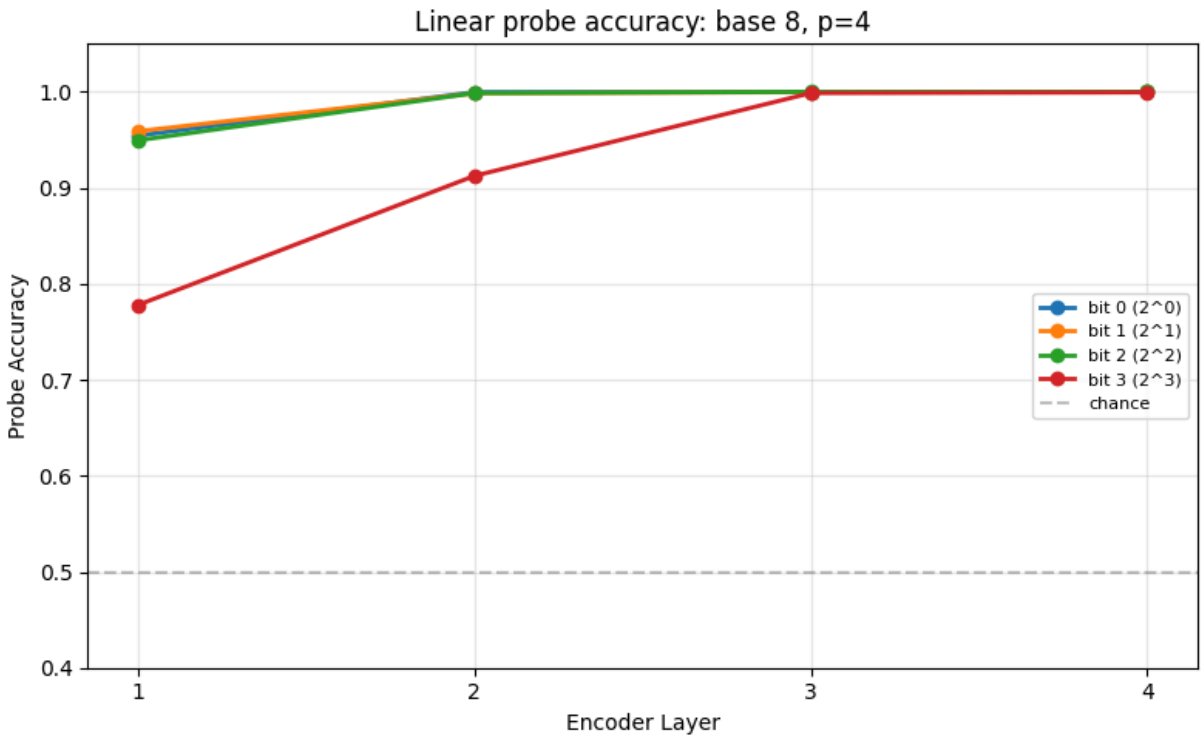



```
bit 0: layers = ['0.523', '0.710', '0.937', '1.000']
bit 1: layers = ['0.498', '0.502', '0.500', '0.510']
bit 2: layers = ['0.509', '0.517', '0.511', '0.508']
bit 3: layers = ['0.489', '0.507', '0.509', '0.496']
```

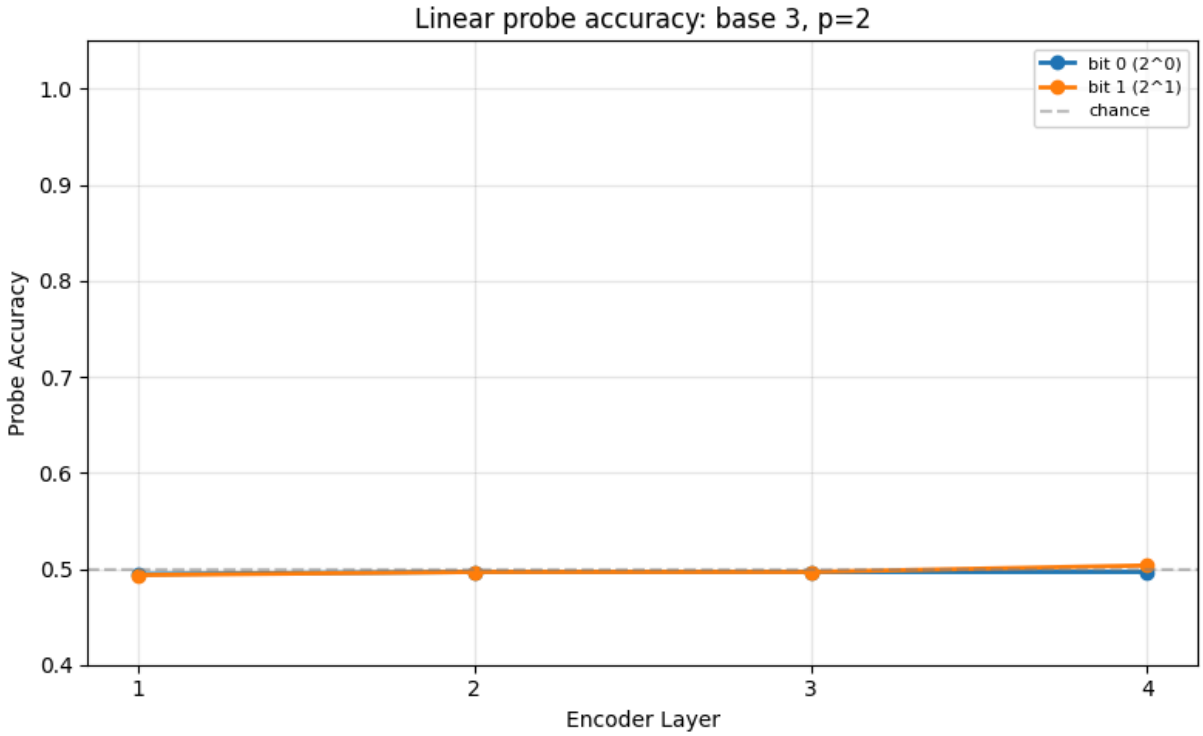
```
Probing B=10, p=4...
```



```
  bit 0: layers = ['0.993', '1.000', '1.000', '1.000']
  bit 1: layers = ['0.625', '0.820', '0.981', '0.999']
  bit 2: layers = ['0.511', '0.558', '0.724', '0.993']
  bit 3: layers = ['0.496', '0.592', '0.715', '0.875']

Probing B=8, p=4...
```

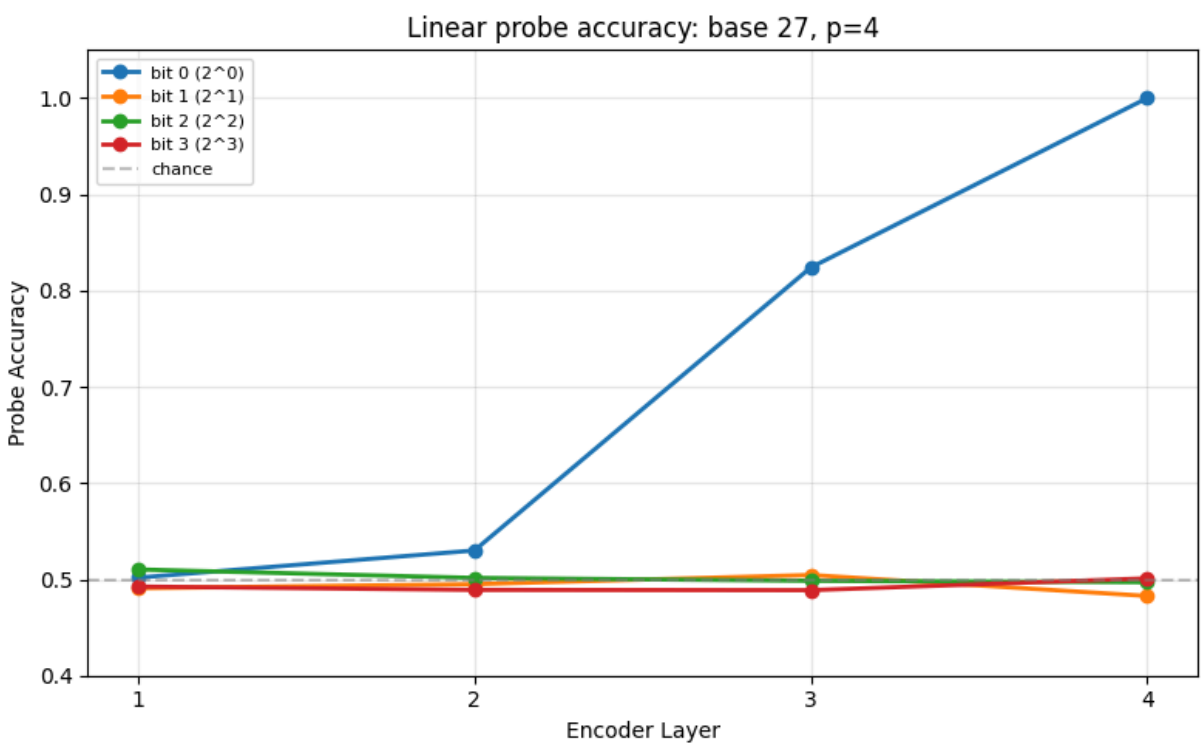



```
  bit 0: layers = ['0.955', '1.000', '1.000', '1.000']
  bit 1: layers = ['0.959', '0.999', '1.000', '1.000']
  bit 2: layers = ['0.950', '0.999', '1.000', '1.000']
  bit 3: layers = ['0.778', '0.913', '0.999', '1.000']

Probing B=3, p=2...
```




```
bit 0: layers = ['0.495', '0.497', '0.497', '0.497']
bit 1: layers = ['0.494', '0.497', '0.497', '0.504']
```

```
Probing B=27, p=4...
```




```
bit 0: layers = ['0.502', '0.530', '0.824', '1.000']
bit 1: layers = ['0.491', '0.495', '0.505', '0.483']
bit 2: layers = ['0.510', '0.502', '0.499', '0.497']
bit 3: layers = ['0.493', '0.489', '0.489', '0.501']
```

14. Ablation: Output in Base 2

Does encoding the *output* in binary (while keeping the input in base B) help? This tests whether the difficulty is in reading base-B inputs or writing base-B outputs.

Since we use classification (output is a class index), the output encoding is already "neutral" — but we can compare to a seq2seq variant where the decoder must produce the answer as a sequence of base-B digits vs binary digits. For now, our classification setup already isolates the input-side difficulty.

In [18]:
```
# The classification formulation already gives us the ablation implicitly:
# the output is always a class label, so there's no output-encoding bottleneck.
# If accuracy is low for odd bases under classification, the difficulty is purely
# on the input/encoding side.

print('Output encoding ablation note:')
print('Since we use classification (not seq2seq), the output is always a class inde
print('Any accuracy differences between bases are purely due to INPUT encoding diff
print()
print('Accuracy summary by base type:')
for p in [1, 4, 8]:
    print(f'\n  p={p}:')
    for b in sorted(BASES):
        if (b, p) in all_results:
            acc = all_results[(b, p)].get('final_acc', '?')
            btype = 'pow2' if b & (b-1) == 0 else ('even' if b % 2 == 0 else 'odd')
            if isinstance(acc, float):
                print(f'    B={b:>2} ({btype:>4}, v₂={nu_2(b)}): {acc:.4f}')
```

```
Output encoding ablation note:
Since we use classification (not seq2seq), the output is always a class index.
Any accuracy differences between bases are purely due to INPUT encoding difficulty.

Accuracy summary by base type:

  p=1:
    B= 2 (pow2, v₂=1): 0.9998
    B= 3 ( odd, v₂=0): 0.9989
    B= 5 ( odd, v₂=0): 0.9993
    B= 8 (pow2, v₂=3): 0.9992
    B=10 (even, v₂=1): 0.9997
    B=16 (pow2, v₂=4): 0.9994
    B=27 ( odd, v₂=0): 0.9991

  p=4:
    B= 2 (pow2, v₂=1): 0.9988
    B= 3 ( odd, v₂=0): 0.0686
    B= 5 ( odd, v₂=0): 0.1270
    B= 8 (pow2, v₂=3): 0.9996
    B=10 (even, v₂=1): 0.9467
    B=16 (pow2, v₂=4): 0.9997
    B=27 ( odd, v₂=0): 0.1279

  p=8:
    B= 2 (pow2, v₂=1): 0.9978
    B= 3 ( odd, v₂=0): 0.0034
    B= 5 ( odd, v₂=0): 0.0041
    B= 8 (pow2, v₂=3): 0.9996
    B=10 (even, v₂=1): 0.0607
    B=16 (pow2, v₂=4): 0.9990
    B=27 ( odd, v₂=0): 0.0051
```

15. Summary Table & Export

In [19]:

```python
# Print a comprehensive summary table
print(f'{"Base":>6} {"v₂":>4}', end='')
for p in P_VALUES:
    print(f'  p={p:>2}', end='')
print()
print('-' * (10 + 6 * len(P_VALUES)))

display_bases = sorted(BASES, key=lambda b: (-nu_2(b), b))
for b in display_bases:
    print(f'{b:>6} {nu_2(b):>4}', end='')
    for p in P_VALUES:
        if (b, p) in all_results:
            acc = all_results[(b, p)].get('final_acc', None)
            if isinstance(acc, float):
                print(f'  {acc:.2f}', end='')
            else:
                print(f'    --', end='')
        else:
            print(f'    --', end='')
    print()
```

```
print(f'\nResults saved in: {SAVE_DIR}')
print('Files per experiment: *_results.json, *_model.pt, *_checkpoint.pt')
print('Figures: accuracy_heatmap.png, learning_curves_p*.png, accuracy_vs_nu2.png')
print('         per_class_B*_p*.png, probes_B*_p*.png')
```

```
  Base   v₂  p= 1  p= 2  p= 4  p= 6  p= 8
----------------------------------------
    16    4  1.00  1.00  1.00  1.00  1.00
     8    3  1.00  1.00  1.00  1.00  1.00
     2    1  1.00  1.00  1.00  1.00  1.00
    10    1  1.00  1.00  0.95  0.24  0.06
     3    0  1.00  0.25  0.07  0.02  0.00
     5    0  1.00  0.51  0.13  0.03  0.00
    27    0  1.00  0.50  0.13  0.02  0.01

Results saved in: /content/drive/MyDrive/binary_suffix_experiment
Files per experiment: *_results.json, *_model.pt, *_checkpoint.pt
Figures: accuracy_heatmap.png, learning_curves_p*.png, accuracy_vs_nu2.png
         per_class_B*_p*.png, probes_B*_p*.png
```

16. Quick Diagnostic: Check for Step-Like Patterns

For each odd-base experiment, check whether the per-class accuracy shows the characteristic all-or-nothing pattern from the Charton paper.

In [20]:

```
def analyze_step_pattern(results, base, p):
    """Check if per-class accuracy shows step-like (all-or-nothing) pattern."""
    key = (base, p)
    if key not in results:
        return None

    r = results[key]
    class_acc = r.get('final_class_acc', {})
    if not class_acc:
        return None

    num_classes = 2 ** p
    accs = [class_acc.get(str(c), 0) for c in range(num_classes)]

    learned = [c for c, a in enumerate(accs) if a > 0.95]
    not_learned = [c for c, a in enumerate(accs) if a < 0.05]
    partial = [c for c, a in enumerate(accs) if 0.05 <= a <= 0.95]

    is_step_like = len(partial) <= max(2, num_classes * 0.1)  # less than 10% parti

    return {
        'learned': len(learned),
        'not_learned': len(not_learned),
        'partial': len(partial),
        'is_step_like': is_step_like,
        'learned_fraction': len(learned) / num_classes,
        'learned_classes': learned,
    }
```

```
print(f'{"Base":>6} {"p":>3} {"v₂":>4} {"Acc":>7} {"Learned":>8} {"NotL":>6} {"Part
print('-' * 55)

for b in sorted(BASES):
    for p in P_VALUES:
        if (b, p) in all_results:
            info = analyze_step_pattern(all_results, b, p)
            if info is not None:
                acc = all_results[(b, p)].get('final_acc', 0)
                print(f'{b:>6} {p:>3} {nu_2(b):>4} {acc:>7.3f} '
                      f'{info["learned"]:>8} {info["not_learned"]:>6} '
                      f'{info["partial"]:>8} {"YES" if info["is_step_like"] else "n
```

```
  Base   p   v₂     Acc  Learned   NotL  Partial  Step?
-------------------------------------------------------
     2   1    1   1.000        2      0        0    YES
     2   2    1   0.999        4      0        0    YES
     2   4    1   0.999       16      0        0    YES
     2   6    1   0.998       64      0        0    YES
     2   8    1   0.998      256      0        0    YES
     3   1    0   0.999        2      0        0    YES
     3   2    0   0.254        1      3        0    YES
     3   4    0   0.069        0     14        2    YES
     3   6    0   0.016        1     63        0    YES
     3   8    0   0.003        0    254        2    YES
     5   1    0   0.999        2      0        0    YES
     5   2    0   0.506        0      0        4     no
     5   4    0   0.127        0     10        6     no
     5   6    0   0.032        0     57        7     no
     5   8    0   0.004        0    252        4    YES
     8   1    3   0.999        2      0        0    YES
     8   2    3   1.000        4      0        0    YES
     8   4    3   1.000       16      0        0    YES
     8   6    3   1.000       64      0        0    YES
     8   8    3   1.000      256      0        0    YES
    10   1    1   1.000        2      0        0    YES
    10   2    1   0.999        4      0        0    YES
    10   4    1   0.947        8      0        8     no
    10   6    1   0.240        1     26       37     no
    10   8    1   0.061        0    203       53     no
    16   1    4   0.999        2      0        0    YES
    16   2    4   0.999        4      0        0    YES
    16   4    4   1.000       16      0        0    YES
    16   6    4   0.999       64      0        0    YES
    16   8    4   0.999      256      0        0    YES
    27   1    0   0.999        2      0        0    YES
    27   2    0   0.505        0      0        4     no
    27   4    0   0.128        0      5       11     no
    27   6    0   0.016        0     60        4    YES
    27   8    0   0.005        0    256        0    YES
```